

Q1: A cord is wound around the circumference of wheel of radius r . The axis of the wheel is horizontal and the moment of inertia about it is I . A weight mg is attached to the cord at the end. The weight falls from rest. After falling through a distance ' h ', the square of angular velocity of wheel will be

(A) $\frac{2mgh}{I+2mr^2}$

(B) $\frac{2gh}{I+mr^2}$

(C) $2gh$

(D) $\frac{2mgh}{I+mr^2}$

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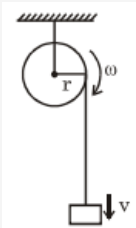
(C) $2gh$

(D) $\frac{2mgh}{I+mr^2}$

Solution:

$$mgh = \frac{1}{2}I\omega^2 + \frac{1}{2}mv^2$$

$$v = \omega r$$



$$mgh = \frac{1}{2}I\omega^2 + \frac{1}{2}m\omega^2 r^2$$

$$\frac{2mgh}{(I+mr^2)} = \omega^2$$

Q2: The trajectory of a projectile in a vertical plane is $y = \alpha x - \beta x^2$ where α and β are constants and x & y are respectively the horizontal and vertical distances of the projectile from the point of projection. The angle of projection θ and the maximum height attained H are respectively given by

(A) $\tan^{-1} \alpha, \frac{4\alpha^2}{\beta}$

(B) $\tan^{-1} \alpha, \frac{\alpha^2}{4\beta}$

(C) $\tan^{-1} \left(\frac{\beta}{\alpha} \right), \frac{\alpha^2}{\beta}$

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(D) $\tan^{-1} \beta, \frac{\alpha^2}{2\beta}$

Solution:

$$y = \alpha x - \beta x^2$$

Comparing with trajectory equation

$$y = x \tan \theta - \frac{1}{2} \frac{gx^2}{u^2 \cos^2 \theta}$$

$$\tan \theta = \alpha \Rightarrow \theta = \tan^{-1} \alpha$$

$$\beta = \frac{1}{2} \frac{g}{u^2 \cos^2 \theta}$$

$$u^2 = \frac{g}{2\beta \cos^2 \theta}$$

Maximum height: H

$$H = \frac{u^2 \sin^2 \theta}{2g} = \frac{g}{2\beta \cos^2 \theta} \frac{\sin^2 \theta}{2g}$$

$$H = \frac{\tan^2 \theta}{4\beta} = \frac{\alpha^2}{4\beta}$$

Q3: Given below are two statements:

Statement I: A second's pendulum has a time period of 1 second.

Statement II: It takes precisely one second to move between the two extreme positions.

In the light of the above statements, chose the correct answer from the options given below

- (A) Statement I is false but Statement II is true
- (B) Statement I is true but Statement II is false
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Solution:

Second pendulum has a time period of 2 sec so statement I is false but from one extreme to other it takes only half the time period so statement II is true.

Q4: A wire of $1\ \Omega$ has a length of 1m. It is stretched till its length increase by 25%. The percentage change in resistance to the nearest integer is

- (A) 76%
- (B) 25%
- (C) 56%
- (D) 12.5%

Q4: A wire of $1\ \Omega$ has a length of 1m. It is stretched till its length increase by 25%. The percentage change in resistance to the nearest integer is

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- (D) 12.5%

Solution:

$$R_0 = 1\ \Omega \quad R_1 = ?$$

$$\ell_0 = 1\text{m} \quad \ell_1 = 1.25\text{m}$$

$$A_0 = A$$

As volume of wire remains constant so

$$A_0 \ell_0 = A_1 \ell_1 \Rightarrow A_1 = \frac{\ell_0 A_0}{\ell_1}$$

Now

$$\text{Resistance (R)} = \frac{\rho \ell}{A}$$

$$\frac{R_0}{R_1} = \frac{\ell_0 / A_0}{\ell_1 / A_1}$$

$$\frac{1}{R_1} = \frac{\ell_0}{A_0} \left(\frac{\ell_0 A_0}{\ell_1 \times \ell_1} \right) R_1 = \frac{\ell_1^2}{\ell_0^2} = 1.5625\ \Omega$$

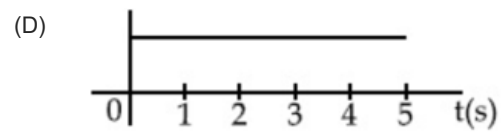
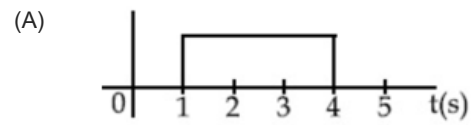
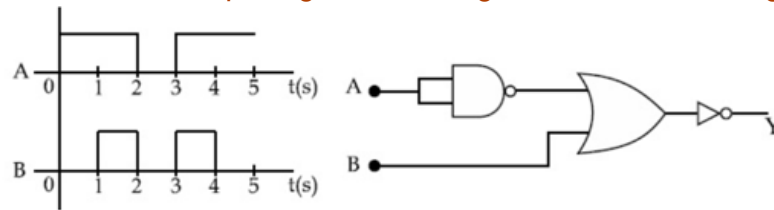
So % change in resistance

$$= \frac{R_1 - R_0}{R_0} \times 100\%$$

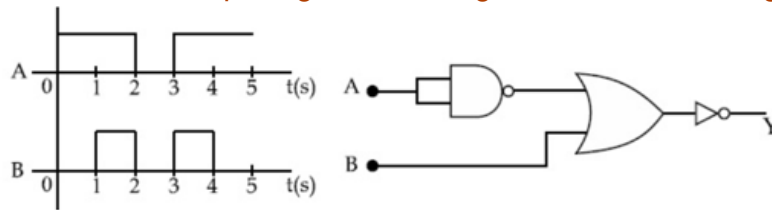
$$= \frac{1.5625 - 1}{1} \times 100\%$$

$$= 56.25\%$$

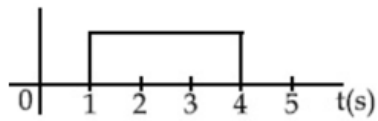
Q5: Draw the output signal Y in the given combination of gates.



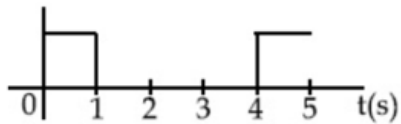
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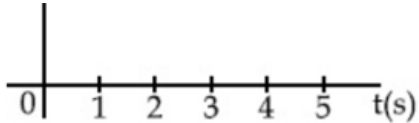
(A)



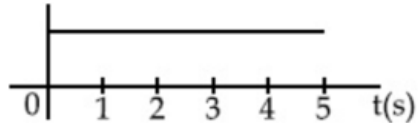
(B)



(C)



(D)



Solution:

According to gates

By Demorgan's law

$$\overline{A + B} = \overline{A} \cdot \overline{B}$$

Thus (b) satisfies the above relation.

Q6: The length of metallic wire is l_1 when tension in it is T_1 . It is l_2 when the tension is T_2 . The original length of the wire will be (Assume that Hooke's law is valid)

(A) $\frac{T_2 l_1 - T_1 l_2}{T_2 - T_1}$

(B) $\frac{l_1 + l_2}{2}$

(C) $\frac{T_1 l_1 - T_2 l_2}{T_2 - T_1}$

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(D) $\frac{T_2 l_1 - T_2 l_2}{T_1 - T_2}$

Solution:

Assuming Hooke's law to be valid,

$$T \propto (\Delta \ell)$$

$$T = k(\Delta \ell)$$

Let ℓ_0 = nature length (original length)

$$\Rightarrow T = k(\ell - \ell_0)$$

$$\text{So, } T_1 = k(\ell_1 - \ell_0) \text{ \& } T_2 = k(\ell_2 - \ell_0)$$

$$\Rightarrow \frac{T_1}{T_2} = \frac{\ell_1 - \ell_0}{\ell_2 - \ell_0}$$

$$\Rightarrow \ell_0 = \frac{T_2 \ell_1 - T_1 \ell_2}{T_2 - T_1}$$

Q7: Given below are two statements:

Statement I: An electric dipole is placed at the centre of a hollow sphere. The flux of electric field through the sphere is zero but the electric field is not zero anywhere in the sphere.

Statement II: If R is the radius of a solid metallic sphere and Q be the total charge on it. The electric field at any point on the spherical surface of radius r (In the light of the above statements, choose the correct answer from the options give below

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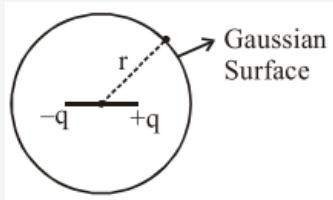
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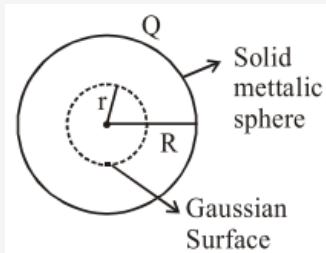


$$\oint \vec{E} \cdot d\vec{s} = \frac{q_{in}}{\epsilon_0} = 0 = \phi$$

Flux of $\vec{E}$ through sphere is zero

But $\oint \vec{E} \cdot d\vec{s} = 0 \Rightarrow \{ \vec{E} \cdot d\vec{s} \neq 0 \}$ for small section ds only

Statement – 2



As charge enclosed within Gaussian surface is equal to zero

$$\phi = \oint \vec{E} \cdot d\vec{s} = 0$$

Option (2) statement -1 correct statement – 2 false

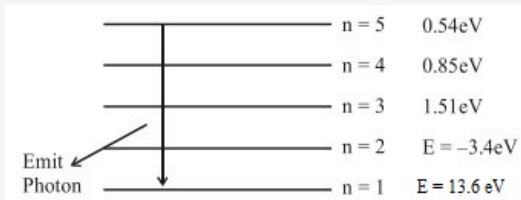
Q8: The recoil speed of a hydrogen atom after it emits a photon in going from $n = 5$ state to $n = 1$ state will be

- (A) 2.19 m/s
- (B) 3.25 m/s
- (C) 4.34 m/s
- (D) 4.17 m/s

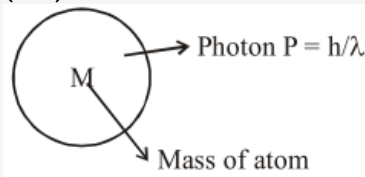
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Solution:



(ΔE) Releases when photon going from $n = 5$ to $n = 1$, $\Delta E = (13.6 - 0.54) \text{ eV} = 13.06 \text{ eV}$



$P_i = P_f$ (By linear momentum conservation)

$$0 = \frac{h}{\lambda} - MV_{\text{Recoil}}$$

$$V_{\text{Recoil}} = \frac{h}{\lambda M}$$

$$\Delta E = \frac{hc}{\lambda} = \frac{hc}{\lambda M} \times M \Rightarrow McV_{\text{Recoil}}$$

$$V_{\text{Recoil}} = \frac{\Delta E}{Mc} = \frac{13.06 \times 1.6 \times 10^{-19}}{1.67 \times 10^{-27} \times 3 \times 10^8} = 4.17 \text{ m/sec}$$

Q9: A tuning fork A of unknown frequency produces 5 beats/s with a fork of known frequency 340 Hz. When fork A is filed, the beat frequency decreases to 2 beats/s. What is the frequency of fork A?

- (A) 338 Hz
- (B) 335 Hz
- (C) 345 Hz
- (D) 342 Hz

Q9: A tuning fork A of unknown frequency produces 5 beats/s with a fork of known frequency 340 Hz. When fork A is filed, the beat frequency decreases to 2 beats/s. What is the frequency of fork A?

- (A) 338 Hz
- ☒ (B) 335 Hz
- (C) 345 Hz
- (D) 342 Hz

Solution:

Initially beat frequency = 5Hz

So, $f_A = 340 \pm 5 = 345\text{Hz}$, 335 Hz

After filing frequency increases slightly

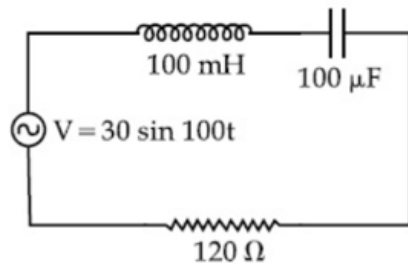
So, new value of frequency of $> f_A$

Now, beat frequency = 2Hz

$\Rightarrow \text{new } f_A = 340 \pm 2 = 342\text{Hz}$, or 338 Hz

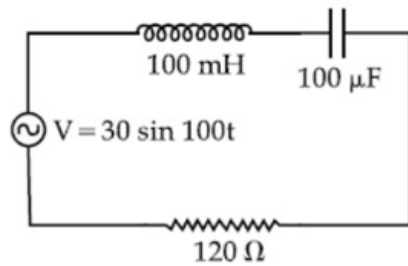
Hence, original frequency of A is $f_A = 335\text{Hz}$

Q10: Find the peak current and resonant frequency of the following circuit (as shown in figure).



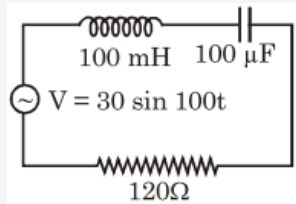
- (A) 2 A and 100 Hz
- (B) 0.2 A and 50 Hz
- (C) 0.2 A and 100 Hz
- (D) 2 A and 50 Hz

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Solution:



$$\text{as given } z = \sqrt{(X_L - X_C)^2 + R^2}$$

$$x_L = \omega L = 100 \times 100 \times 10^{-3} = 10\Omega$$

$$x_C = \frac{1}{\omega C} = \frac{1}{100 \times 100 \times 10^{-6}} = 100\Omega$$

$$z = \sqrt{(10 - 100)^2 + 120^2} = \sqrt{90^2 + 120^2}$$

$$z = 150\Omega$$

$$i_{\text{peak}} = \frac{\Delta v}{z} = \frac{30}{150} = \frac{1}{5} \text{ A} = 0.2 \text{ A}$$

& For resonant frequency

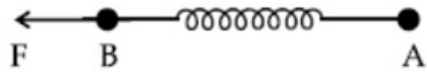
$$\Rightarrow \omega L = \frac{1}{\omega C} \Rightarrow \omega^2 = \frac{1}{LC} \Rightarrow \omega = \frac{1}{\sqrt{LC}}$$

$$\& f = \frac{1}{2\pi\sqrt{LC}} \Rightarrow \frac{1}{2\pi\sqrt{100 \times 10^{-3} \times 100 \times 10^{-6}}}$$

$$= \frac{100\sqrt{10}}{2\pi} = \frac{100\pi}{2\pi} = 50 \text{ Hz}$$

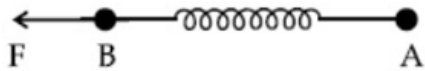
$$\text{As } \sqrt{10} \approx \pi$$

Q11: Two masses A and B, each of mass M are fixed together by a massless spring. A force acts on the mass B as shown in figure. If the mass A starts moving away from mass B with magnitude of acceleration as ' a ', then the magnitude of acceleration of mass B at that instant will be:



- (A) $\frac{MF}{F+Ma}$
- (B) $\frac{Ma-F}{M}$
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(B) $\frac{Ma-F}{M}$

(C) $\frac{F+Ma}{M}$

(D) $\frac{F-Ma}{M}$

Solution:

$$a_{cm} = \frac{m_1 a_1 + m_2 a_2}{m_1 + m_2}$$

$$-\frac{F}{2M} = \frac{Ma + Ma_B}{2M} \text{ (consider direction of movement of A to be positive)}$$

$$a_B = -\frac{F+Ma}{M} \text{ (ignore negative sign for magnitude)}$$

Q12: The incident ray, reflected ray and the outward drawn normal are denoted by the unit vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ -respectively. Then choose the correct relation for these vectors.

(A) $\vec{b} = \vec{a} + 2\vec{c}$

(B) $\vec{b} = \vec{a} - 2(\vec{a} \cdot \vec{c})\vec{c}$

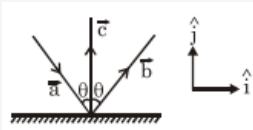
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Solution:



$$\vec{a} = \sin \theta \hat{i} - \cos \theta \hat{j}$$

$$\vec{b} = \sin \theta \hat{i} + \cos \theta \hat{j}$$

$$\vec{c} = \hat{j}$$

$$\vec{a} - 2(\vec{a} \cdot \vec{c})\vec{c} = \sin \theta \hat{i} + \cos \theta \hat{j}$$

Q13: A scooter accelerates from rest for time t_1 at constant rate a_1 and then retards at constant rate a_2 for time t_2 and comes to rest. The correct value of $\frac{t_1}{t_2}$ will be

(A) $\frac{a_1 + a_2}{a_1}$

(B) $\frac{a_1}{a_2}$

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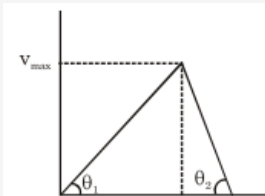
(B) $\frac{a_1}{a_2}$

(C) $\frac{a_2}{a_1}$

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Solution:

Draw vt curve



$$\tan \theta_1 = a_1 = \frac{v_{\max}}{t_1}$$

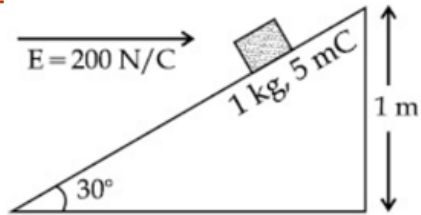
$$\& \tan \theta_2 = a_2 = \frac{v_{\max}}{t_2}$$

+ above

$$\frac{t_1}{t_2} = \frac{a_2}{a_1}$$

Q14: An inclined plane making an angle of 30° with the horizontal is placed in a uniform horizontal electric field $200 \frac{N}{C}$ as shown in the figure. A body of mass 1 kg and charge 5mC is allowed to slide down from rest at a height of 1m. If the coefficient of friction is 0.2, find the time taken by the body to reach the bottom.

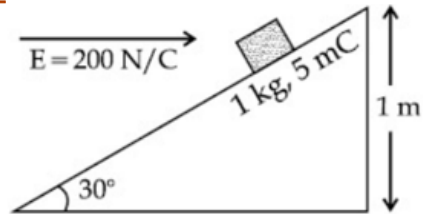
$$\left[g = 9.8 \text{ m/s}^2; \sin 30^\circ = \frac{1}{2}; \cos 30^\circ = \frac{\sqrt{3}}{2} \right]$$



- (A) 2.3 s
- (B) 1.3 s
- (C) 0.92 s
- (D) 0.46 s

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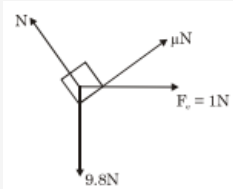
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Solution:

FBD



Here $N = 9.8 \cos 30 + 1 \sin 30 \approx 9\text{N}$

$$\text{So } a = \frac{9.8 \sin 30 - 1 \cos 30 - \mu N}{1}$$

$$a = 2.233 \text{ m/s}^2$$

$$\text{By } S = ut + \frac{1}{2}at^2$$

$$\frac{1}{\sin 30^\circ} = \frac{1}{2}(2.233)t^2$$

$$t \approx 1.3 \text{ sec}$$

Q15: An aeroplane, with its wings spread 10 m, is flying at a speed of 180 km/h in a horizontal direction. The total intensity of earth's field at that part is $25 \times 10^{-4} \text{ Wb/m}^2$ and the angle of dip is 60° . The emf induced between the tips of the plane wings will be:

- (A) 62.50 mV
- (B) 54.125 mV
- (C) 88.37 mV
- (D) 108.25 mV

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Solution:

$$\begin{aligned}\epsilon &= BVL \sin \theta \\ &= (2.5 \times 10^{-4} T) \left(180 \times \frac{5}{18} m/s\right) (10m) \sin 60^\circ \\ &= 108.25 \times 10^{-3} V\end{aligned}$$

Q16: Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: For a simple microscope, the angular size of the object equals the angular size of the image.

Reason R: Magnification is achieved as the small object can be kept much closer to the eye than 25 cm and hence it subtends a large angle.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) A is true but R is false
- (B) Both A and R are true but R is not the correct explanation of A
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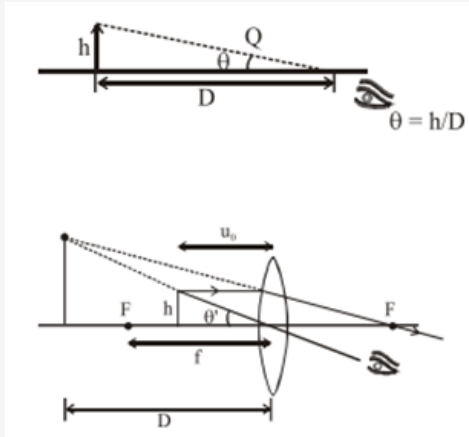
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- ☒ (D) Both A and R are true and R is the correct explanation of A

Solution:



$\theta = \frac{h}{u_0}$; θ' is same for both object and image

$$m = \frac{\theta'}{\theta} = \frac{D}{u_0}$$

$$u_0 < D$$

Hence $m > 1$

Q17: A particle executes S.H.M., the graph of velocity as a function of displacement is

- (A) A circle
- (B) An ellipse
- (C) A parabola
- (D) A helix

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Solution:

$$v^2 = \omega^2 (A^2 - x^2)$$

$$\frac{v^2}{\omega^2} + x^2 = A^2$$

$$\frac{v^2}{(\omega A)^2} + \frac{x^2}{A^2} = 1$$

This is an equation of an ellipse

Q18: If 'C' and 'V' represent capacity and voltage respectively then what are the dimensions of λ where $C/V = \lambda$?

(A) $[M^{-1} L^{-3} I^{-2} T^{-7}]$

(B) $[M^{-3} L^{-4} I^3 T^7]$

(C) $[M^{-2} L^{-4} I^3 T^7]$

(D) $[M^{-2} L^{-3} I^2 T^6]$

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(D) $[M^{-2} L^{-3} I^2 T^6]$

Solution:

$$\lambda = \frac{C}{V} = \frac{Q/V}{V} = \frac{Q}{V^2}$$

$$V = \frac{\text{work}}{Q}$$

$$\lambda = \frac{Q^3}{(\text{work})^2} = \frac{(It)^3}{(Fs)^2}$$

$$= \frac{[I^3 T^3]}{[ML^2 T^{-2}]^2} = [M^{-2} L^{-4} I^3 T^7]$$

Q19: The internal energy (U), pressure (P) and volume (V) of an ideal gas are related as $U = 3PV + 4$. The gas is

- (A) Diatomic only
- (B) Polyatomic only
- (C) Either monoatomic or diatomic
- (D) Monoatomic only

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- (A) Diatomic only
- ☒ (B) Polyatomic only
- (C) Either monoatomic or diatomic
- (D) Monoatomic only

Solution:

$$U = 3PV + 4$$

$$\frac{nf}{2}RT = 3PV + 4$$

$$\frac{f}{2}PV = 3PV + 4$$

$$f = 6 + \frac{8}{PV}$$

Since degree of freedom is more than 6 therefore gas is polyatomic.

Q20: A radioactive sample is undergoing α decay. At any time t_1 , its activity is A and another time t_2 , the activity is $\frac{A}{5}$. What is the average life time for the sample?

(A) $\frac{t_2 - t_1}{\ln 5}$

(B) $\frac{t_1 - t_2}{\ln 5}$

(C) $\frac{\ln 5}{t_2 - t_1}$

(D) $\frac{\ln(t_2 + t_1)}{2}$

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(D) $\frac{\ln(t_2 + t_1)}{2}$

Solution:

Let initial activity be A_0

$$A = A_0 e^{-\lambda t_1} \dots (i)$$

$$\frac{A}{5} = A_0 e^{-\lambda t_2} \dots (ii)$$

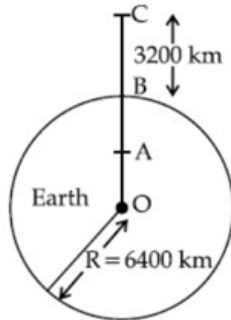
$$(i) \div (ii)$$

$$5 = e^{\lambda(t_2 - t_1)}$$

$$\lambda = \frac{\ln 5}{t_2 - t_1} = \frac{1}{\tau}$$

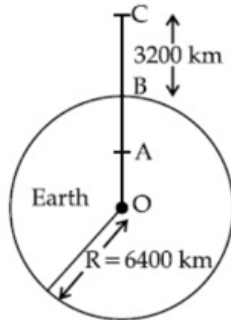
$$\tau = \frac{t_2 - t_1}{\ln 5}$$

Q21: In the reported figure of earth, the value of acceleration due to gravity is same at point A and C but it is smaller than that of its value at point B (surface of the earth). The value of OA : AB will be x : y. The value of x is:



Given 2

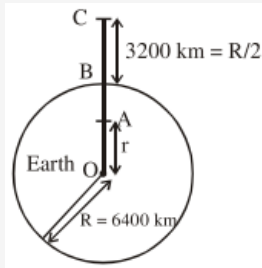
Q21: In the reported figure of earth, the value of acceleration due to gravity is same at point A and C but it is smaller than that of its value at point B (surface of the earth). The value of OA : AB will be x : y. The value of x is:



Given 2

4.00

Solution:



$$g_A = \frac{GM(r)}{R^3}$$

$$g_C = \frac{GM}{\left(R + \frac{R}{2}\right)^2}$$

$$g_A = g_C \quad \Rightarrow \quad \frac{r}{R^3} = \frac{1}{\frac{9}{4}R^2} \quad \Rightarrow \quad r = \frac{4R}{9}$$

$$\text{So } OA = \frac{4R}{9}; AB = R - r = \frac{5R}{9}$$

$$OA:AB=4:5$$

Q22: A particle executes S.H.M. with amplitude 'A' and time period 'T'. The displacement of the particle when its speed is half of maximum speed is $\frac{\sqrt{x}A}{2}$. The value of x is:

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Solution:

$$V = \omega \sqrt{A^2 - x^2} \quad V_{max} = A\omega$$

$$\frac{A\omega}{2} = \omega \sqrt{A^2 - x^2}$$

$$\frac{A^2}{4} = A^2 - x^2$$

$$x^2 = \frac{3A^2}{4}$$

$$x = \frac{\sqrt{3}}{2}A$$

Q23: The volume V of a given mass of monoatomic gas changes with temperature T according to the relation $V = KT^{\frac{2}{3}}$. The work done when temperature changes by 90 K will be xR . The value of x is: (Assume one mole of gas only)
[R = universal gas constant]

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 [R = universal gas constant]

60.00

Solution:

We know that work done is

$$W = \int P dV \dots (1)$$

$$\Rightarrow P = \frac{nRT}{V} \dots (2)$$

$$\Rightarrow W = \int \frac{nRT}{V} dV \dots (3)$$

$$\text{And } V = KT^{2/3} \dots (4)$$

$$\Rightarrow W = \int \frac{nRT}{KT^{2/3}} dV \dots (5)$$

$$\Rightarrow \text{from (4)} : dV = \frac{2}{3} KT^{-1/3} dT$$

$$\Rightarrow W = \int_{T_1}^{T_2} \frac{nRT}{KT^{2/3}} \cdot \frac{2}{3} K \frac{1}{T^{1/3}} dT$$

$$\Rightarrow W = \frac{2}{3} nR \times (T_2 - T_1) \dots (6)$$

$$\Rightarrow T_2 - T_1 = 90K \dots (7)$$

$$\Rightarrow W = \frac{2}{3} nR \times 90$$

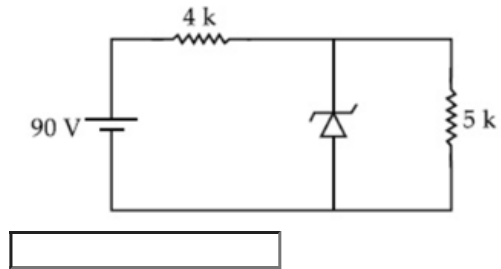
$$\Rightarrow W = 60nR$$

For 1 mole of gas

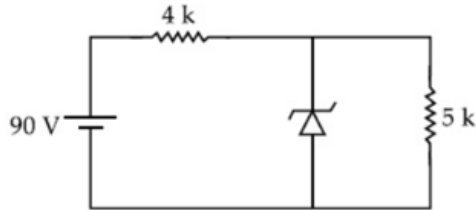
$$n = 1$$

$$\text{So } W = 60 R$$

Q24: The Zener diode is being operated across potential difference $V_z = 30\text{ V}$. The current passing through the diode for the following circuit is _____mA.

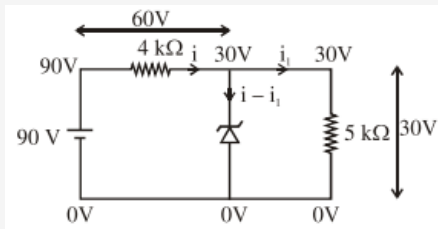


Q24: The Zener diode is being operated across potential difference $V_z = 30\text{ V}$. The current passing through the diode for the following circuit is _____mA.



9.00

Solution:



$$i = \frac{60}{4000} A$$

$$i_1 = \frac{30}{5000} A$$

$$i - i_1 = \frac{60}{4000} - \frac{30}{5000} = \frac{9}{1000} A$$

Current from zener diode

$$i_2 = i - i_1 = 9mA$$

Q25: Time period of a simple pendulum is T . The time taken to complete $\frac{5}{8}$ oscillations starting from mean position is $\frac{\alpha}{\beta}T$. The value of α is:

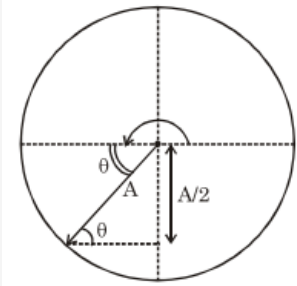
Q25: Time period of a simple pendulum is T . The time taken to complete $\frac{5}{8}$ oscillations starting from mean position is $\frac{\alpha}{\beta}T$. The value of α is:

7.00

Solution:

$\frac{5}{8}$ th of oscillation = $\left(\frac{1}{2} + \frac{1}{8}\right)^{th}$ of oscillation.

We can solve it using phasor diagram.



$$\pi + \theta = \omega t$$

$$\pi + \frac{\pi}{6} = \left(\frac{2\pi}{T}\right) t$$

$$\frac{7\pi}{6} = \left(\frac{2\pi}{T}\right) t$$

$$t = \frac{7T}{12}$$

Q26: 1 mole of rigid diatomic gas performs a work of $\frac{Q}{5}$ when heat Q is supplied to it. The molar heat capacity of the gas during this transformation is $\frac{xR}{8}$. The value of x is:

[R = universal gas constant]

Q26: 1 mole of rigid diatomic gas performs a work of $\frac{Q}{5}$ when heat Q is supplied to it. The molar heat capacity of the gas during this transformation is $\frac{xR}{8}$. The value of x is:

[R = universal gas constant]

25.00

Solution:

$$Q = \Delta U + W$$

$$Q = \Delta U + \frac{Q}{5}$$

$$\Delta U = \frac{4Q}{5}$$

$$nC_V \Delta T = \frac{4}{5} nC \Delta T$$

$$\frac{5}{4} C_V = C$$

$$C = \frac{5}{4} \left(\frac{f}{2} \right) R = \frac{5}{4} \left(\frac{5}{2} \right) R$$

$$C = \frac{25}{8} R$$

$$x = 25$$

Q27: If the highest frequency modulating a carrier is 5 kHz, then the number of AM broadcast stations accommodated in a 90 kHz bandwidth are:

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9.00

Solution:

B.W. (Bandwidth) = $2 \times$ maximum frequency at modulating signal

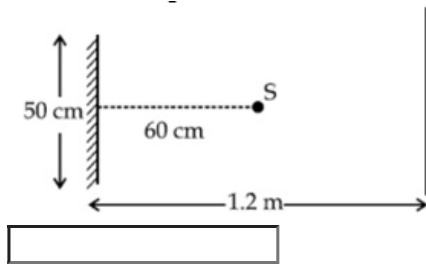
= $2 \times 5\text{kHz}$

= 10 kHz

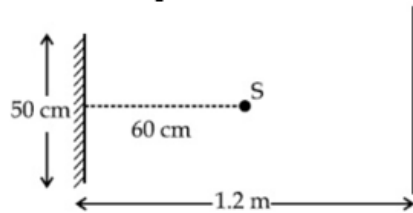
$\therefore$ No of stations accommodated are

$$= \frac{90}{10} = 9$$

Q28: A point source of light S, placed at a distance 60cm in front of the centre of a plane mirror of width 50 cm, hangs vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance 1.2m from it (see in the figure). The distance between the extreme points where he can see the image of the light source in the mirror is _____ cm.

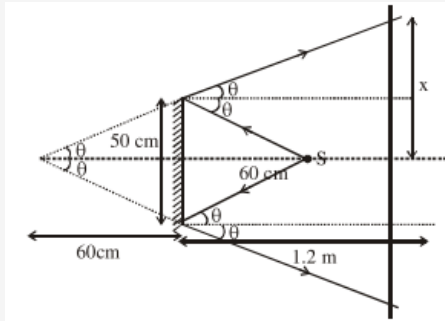


Q28: A point source of light S, placed at a distance 60cm in front of the centre of a plane mirror of width 50 cm, hangs vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance 1.2m from it (see in the figure). The distance between the extreme points where he can see the image of the light source in the mirror is _____cm.



150.00

Solution:



$$\tan \theta = \frac{25}{60} = \frac{x}{180}$$

$$x = 75 \text{ cm}$$

$$\text{So distance between extreme point} = 2x = 2 \times 75 = 150 \text{ cm}$$

Q29: Two stream of photons, possessing energies equal to twice and ten times the work function of metal are incident on the metal surface successively. The value of ratio of maximum velocities of the photoelectrons emitted in the two respective cases is $x : 3$. The value of x is:

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Solution:

$$KE_{\max} = hv - \phi$$

$$\frac{1}{2}mv^2 = hv - \phi$$

$$v = \sqrt{\frac{2(hv - \phi)}{m}}$$

$$\text{Given } hv_1 = 2\phi$$

$$hv_2 = 10\phi$$

$$\therefore \frac{v_1}{v_2} = \sqrt{\frac{hv_1 - \phi}{hv_2 - \phi}}$$

$$\frac{v_1}{v_2} = \sqrt{\frac{2\phi - \phi}{10\phi - \phi}} = \frac{1}{3}$$

Q30: 27 similar drops of mercury are maintained at 10V each. All these spherical drops combine into a single big drop. The potential energy of the bigger drop is _____ times that of a smaller drop.

Q30: 27 similar drops of mercury are maintained at 10V each. All these spherical drops combine into a single big drop. The potential energy of the bigger drop is _____ times that of a smaller drop.

243.00

Solution:

$$(27) \left(\frac{4}{3} \pi r^3 \right) = \frac{4}{3} \pi R^3$$

$$R = 3r$$

Potential energy of smaller drop

$$U_1 = \frac{3}{5} \frac{kq^2}{r}$$

Potential energy of bigger drop

$$U = \frac{3}{5} \frac{k(27q)^2}{R}$$

$$U = \frac{3}{5} k \frac{(27)(27)q^2}{3r}$$

$$U = \frac{(27)(27)}{3} \left(\frac{3}{5} \frac{kq^2}{r} \right)$$

$$U = 243U_1$$

Q31: Match List-I with List-II:

	List I		List II
A	Sucrose	(i)	β -D-Galactose and β -D-Glucose
B	Lactose	(ii)	α -D-Glucose and β -D-Fructose
C	Maltose	(iii)	α -D-Glucose and α -D-Glucose

(A) A – (iii), B – (i), C – (ii)

(B) A – (i), B – (iii), C – (ii)

(C) A – (iii), B – (ii), C – (i)

(D) A – (ii), B – (i), C – (iii)

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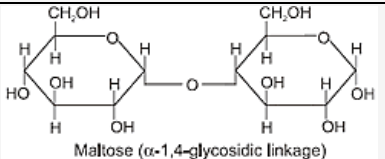
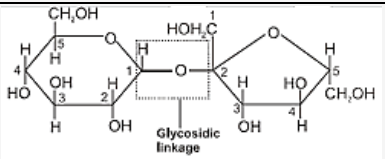
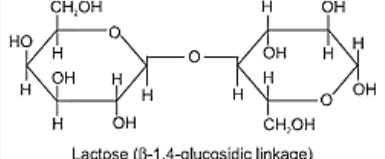
(A) A – (iii), B – (i), C – (ii)

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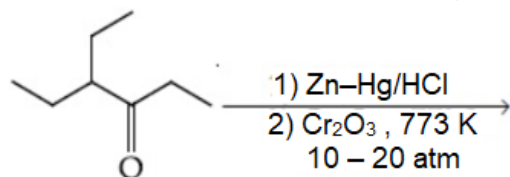
(C) A – (iii), B – (ii), C – (i)

(D) A – (ii), B – (i), C – (iii)

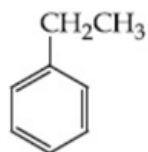
Solution:

Disaccharide	Structure	Monomeric unit linkage
Maltose	 Maltose (α-1,4-glycosidic linkage)	$\alpha - D (+)$ Glucose + $\alpha - D (+)$ Glucose ($\alpha - 1, 4$ -glycosidic linkage)
Sucrose	 Glycosidic linkage	$\alpha - D$ Glucose + $\beta - D$ - Fructose ($\alpha - 1, \beta - 2$ glycosidic linkage)
Lactose	 Lactose (β-1,4-glycosidic linkage)	$\beta - D (+)$ Glucose + $\beta - D (+)$ Glucose ($\beta - 1, 4$ -glycosidic linkage)

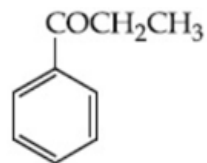
Q32: Consider the below reaction, the major product among the following is



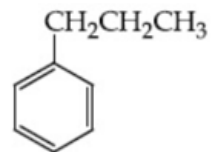
(A)



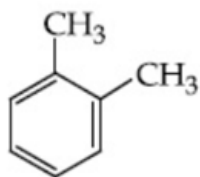
(B)



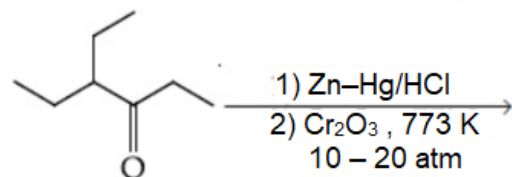
(C)

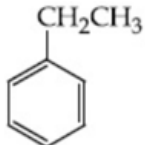
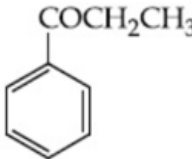
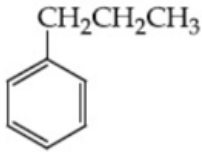
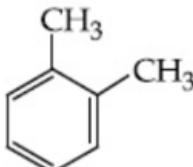


(D)

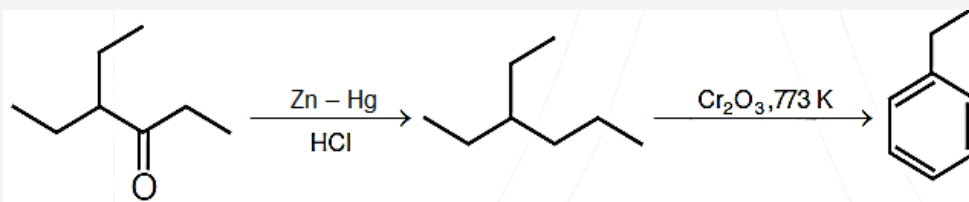


Q32: Consider the below reaction, the major product among the following is



- (A) 
- (B) 
- (C) 
- (D) 

Solution:



Q33: Seliwanoff test and Xanthoproteic test are used for the identification of _____ and _____ respectively.

- (A) Ketoses, aldoses
- (B) Proteins, ketoses
- (C) Aldoses, ketoses
- (D) Ketoses, proteins

Q33: Seliwanoff test and Xanthoproteic test are used for the identification of _____ and _____ respectively.

- (A) Ketoses, aldoses
- (B) Proteins, ketoses
- (C) Aldoses, ketoses
- (D) Ketoses, proteins

Solution:

Test	Reagent	Observations	Reason	Test given By
Seliwanoff's Test	0.5% Resorcinol in conc. HCl and heat for 5 minutes	Fiery red colour or Coloured solution.	Complex formation	Fructose gives fiery red solution but glucose, maltose and sucrose gives brown/violet coloured solutions. (Difference between fructose and glucose)

Xanthoproteic test is used to detect a presence of protein.

Q34: Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: In TlI_3 , isomorphous to CsI_3 , the metal is present in +1 oxidation state.

Reason R: Tl metal has fourteen *f* electrons in its electronic configuration.

In the light of the above statements, choose the most appropriate answer from the options given below

- (A) Both A and R are correct and R is the correct explanation of A
- (B) A is not correct but R is correct
- (C) Both A and R are correct but R is not the correct explanation of A
- (D) A is correct but R is not correct

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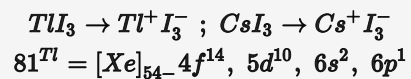
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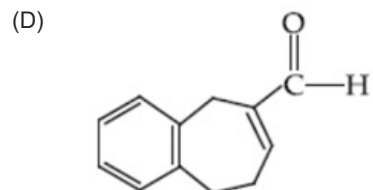
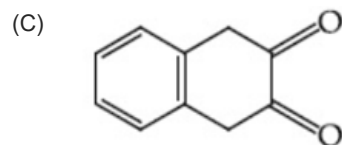
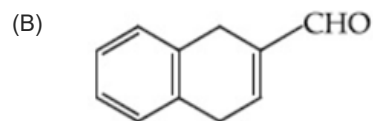
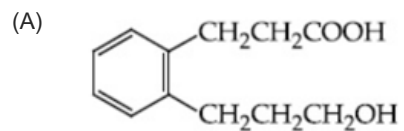
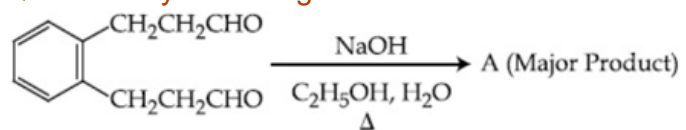
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- (A) Both A and R are correct and R is the correct explanation of A
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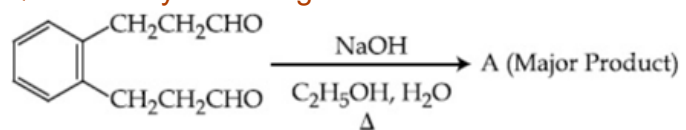
Solution:



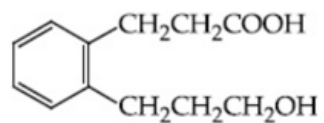
Q35: Identify A in the given chemical reaction.



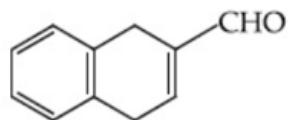
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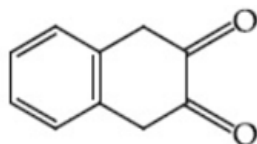
(A)



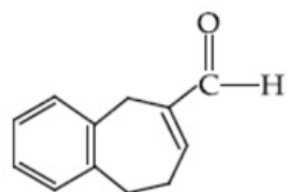
(B)



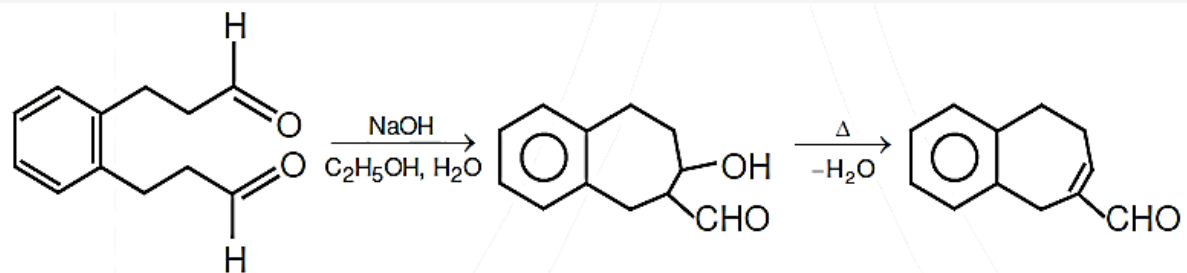
(C)



(D)



Solution:



Q36: The correct order of electron gain enthalpy is

- (A) $S > Se > Te > O$
- (B) $Te > Se > S > O$
- (C) $S > O > Se > Te$
- (D) $O > S > Se > Te$

Q36: The correct order of electron gain enthalpy is

- (A) $S > Se > Te > O$
- (B) $Te > Se > S > O$
- (C) $S > O > Se > Te$
- (D) $O > S > Se > Te$

Solution:

The order of electron gain enthalpy is $S > Se > Te > O$

Q37: Match List-I with List-II

	List I		List II
A	Sodium Carbonate	(i)	Deacon
B	Titanium	(ii)	Castner-Kellner
C	Chlorine	(iii)	Van-Arkel
D	Sodium hydroxide	(iv)	Solvay

Choose the correct answer from the options given below

- (A) A – (iv), B – (i), C – (ii), D – (iii)
- (B) A – (iii), B – (ii), C – (i), D – (iv)
- (C) A – (i), B – (iii), C – (iv), D – (ii)
- (D) A – (iv), B – (iii), C – (i), D – (ii)

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Choose the correct answer from the options given below

- (A) A – (iv), B – (i), C – (ii), D – (iii)
- (B) A – (iii), B – (ii), C – (i), D – (iv)
- (C) A – (i), B – (iii), C – (iv), D – (ii)
- (D) A – (iv), B – (iii), C – (i), D – (ii)

Solution:

	Column – I		Column – II
(a)	Na_2CO_3	(i)	Solvay's process
(b)	Ti	(ii)	Van arkel process
(c)	Cl_2	(iii)	Deacon's process
(d)	NaOH	(iv)	Castner Kellner process

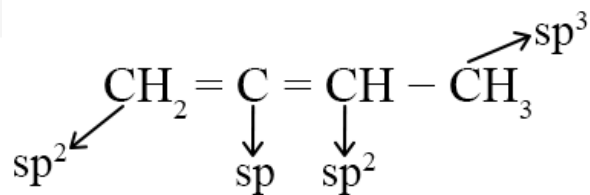
Q38: In $\overset{1}{\text{C}}\text{H}_2 = \overset{2}{\text{C}} = \overset{3}{\text{C}}\text{H} - \overset{4}{\text{C}}\text{H}_3$ molecule, the hybridization of carbon 1, 2, 3 and 4 respectively, are:

- (A) sp^2 , sp, sp^2 , sp^3
- (B) sp^3 , sp, sp^3 , sp^3
- (C) sp^2 , sp^3 , sp^2 , sp^3
- (D) sp^2 , sp^2 , sp^2 , sp^3

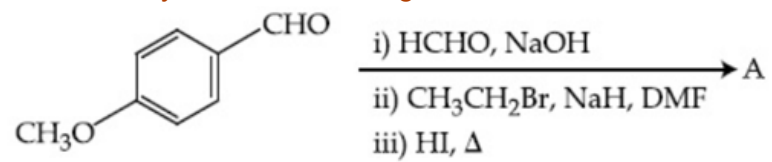
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- (A) sp^2 , sp, sp^2 , sp^3
- (B) sp^3 , sp, sp^3 , sp^3
- (C) sp^2 , sp^3 , sp^2 , sp^3
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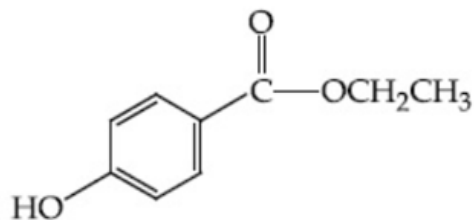
Solution:



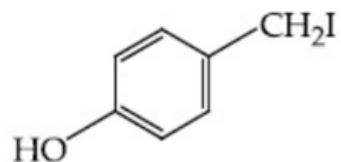
Q39: Identify A in the following chemical reaction.



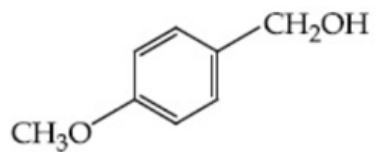
(A)



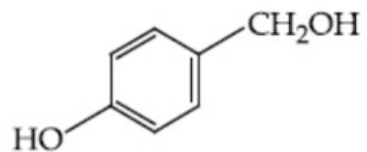
(B)



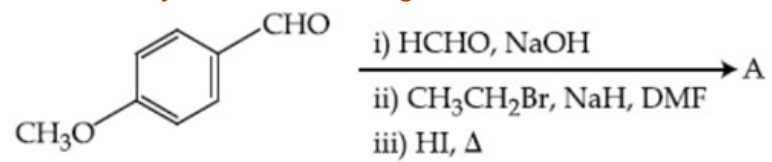
(C)



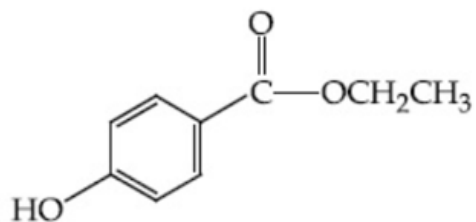
(D)



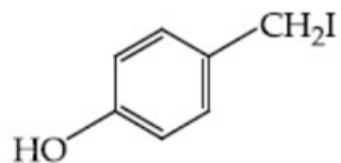
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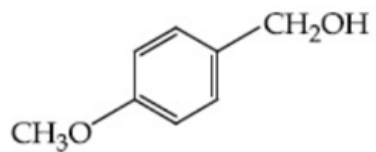
(A)



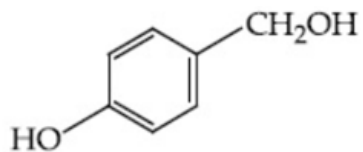
(B)



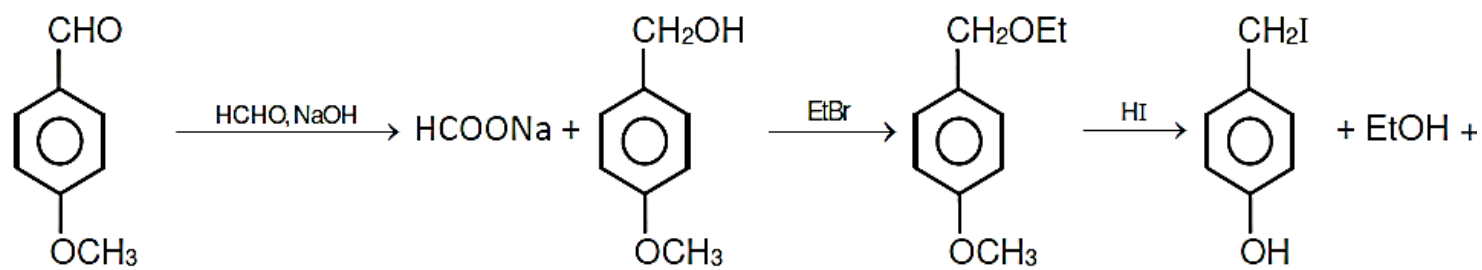
(C)



(D)



Solution:



Q40: 2, 4-DNP test can be used to identify

- (A) Amine
- (B) Aldehyde
- (C) Ether
- (D) Halogens

Q40: 2, 4-DNP test can be used to identify

- (A) Amine
- ☒ (B) Aldehyde
- (C) Ether
- (D) Halogens

Solution:

Carbonyl group (aldehyde & ketone) give positive test with 2,4-DNP.

Q41: Which pair of oxides is acidic in nature?

- (A) N_2O , BaO
- (B) CaO , SiO_2
- (C) B_2O_3 , SiO_2
- (D) B_2O_3 , CaO

Q41: Which pair of oxides is acidic in nature?

(A) N_2O , BaO

(B) CaO , SiO_2

(C) B_2O_3 , SiO_2

(D) B_2O_3 , CaO

Solution:

B_2O_3 & SiO_2 are non metal oxide & hence are acidic in nature

Q42: Ceric ammonium nitrate and $\text{CHCl}_3/\text{alc. KOH}$ are used for the identification of functional groups present in _____ and _____ respectively.

- (A) Alcohol, amine
- (B) Amine, phenol
- (C) Alcohol, phenol
- (D) Amine, alcohol

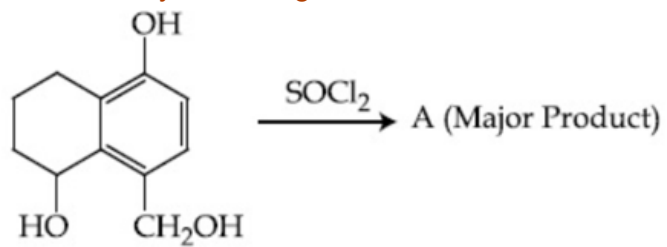
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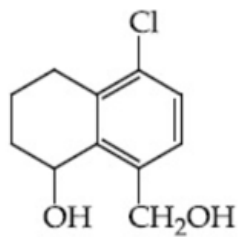
Solution:

Alcohols give the positive test with Ceric ammonium nitrate & 1° amines give the positive test with $\text{CHCl}_3 + \text{KOH}$ i.e. carbylamine test.

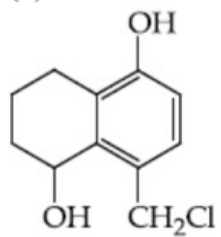
Q43: Identify A in the given reaction.



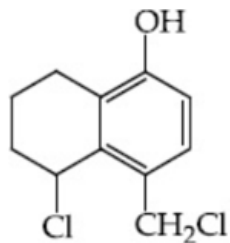
(A)



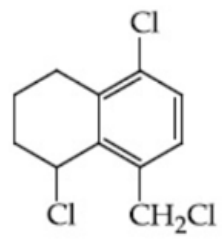
(B)



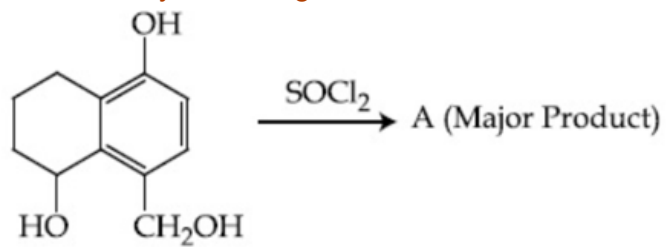
(C)



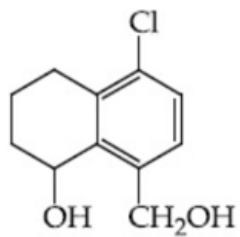
(D)



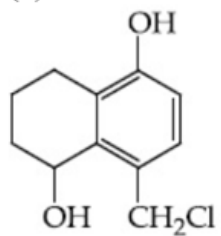
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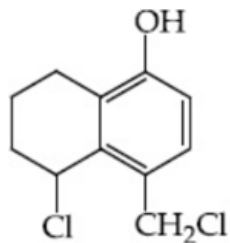
(A)



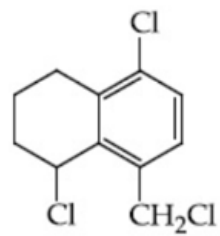
(B)



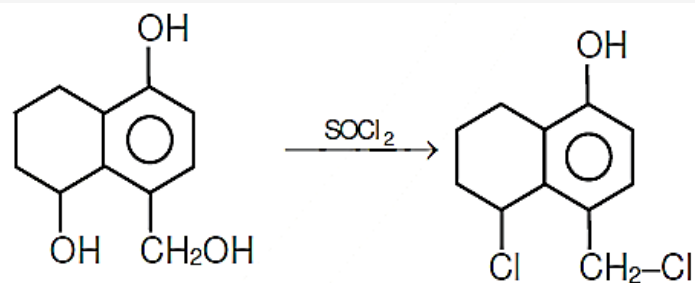
(C)



(D)



Solution:



Q44: Calgon is used for water treatment. Which of the following statement is not true about Calgon?

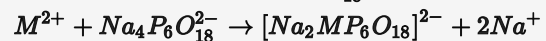
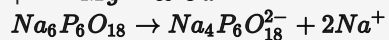
- (A) It is also known as Graham's salt
- (B) Calgon contains the 2nd most abundant element by weight in the Earth's crust
- (C) It is polymeric compound and is water soluble
- (D) It doesnot remove Ca^{2+} ion by precipitation

Q44: Calgon is used for water treatment. Which of the following statement is not true about Calgon?

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- (B) Calgon contains the 2nd most abundant element by weight in the Earth's crust
- (C) It is polymeric compound and is water soluble
- (D) It does not remove Ca^{2+} ion by precipitation

Solution:

- (1) Sodium hexametaphosphate is also known as Graham's salt.
- (2) Calgon is sodium hexametaphosphate ($\text{Na}_6\text{P}_6\text{O}_{18}$) or $(\text{NaPO}_3)_6$
- (3) Fe is second most abundant metal in earth crust. Calgon does not contain it.
- (4) It keeps the Mg^{+2} & Ca^{2+} in solution.



Q45: Match List-I with List-II

	List-I		List-II
A	Siderite	(i)	Cu
B	Calamine	(ii)	Ca
C	Malachite	(iii)	Fe
D	Cryolite	(iv)	Al
		(v)	Zn

Choose the correct answer from the options given below

- (A) A – (i), B – (ii), C – (iii), D – (iv)
- (B) A – (iii), B – (v), C – (i), D – (iv)
- (C) A – (iii), B – (i), C – (v), D – (ii)
- (D) A – (i), B – (ii), C – (v), D – (iii)

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		(v)	Zn

Choose the correct answer from the options given below

(A) A – (i), B – (ii), C – (iii), D – (iv)

(B) A – (iii), B – (v), C – (i), D – (iv)

(C) A – (iii), B – (i), C – (v), D – (ii)

(D) A – (i), B – (ii), C – (v), D – (iii)

Solution:

	Ore	Formula
(a)	Siderite	- $FeCO_3$
(b)	Calamine	- $ZnCO_3$
(c)	Cryolite	- Na_3AlF_6
(d)	Malachite	- $CuCO_3 \cdot Cu(OH)_2$

Q46: Match List-I with List-II

	List-I(Molecule)		List-II(Bond order)
A	Ne ₂	(i)	1
B	N ₂	(ii)	2
C	F ₂	(iii)	0
D	O ₂	(iv)	3

Choose the correct answer from the options given below

- (A) A – (iv), B – (iii), C – (ii), D – (i)
- (B) A – (ii), B – (i), C – (iv), D – (iii)
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Choose the correct answer from the options given below

- (A) A – (iv), B – (iii), C – (ii), D – (i)
- (B) A – (ii), B – (i), C – (iv), D – (iii)
- (C) A – (iii), B – (iv), C – (i), D – (ii)**
- (D) A – (i), B – (ii), C – (iii), D – (iv)

Solution:

-

	Molecule	Bond Order
(a)	Ne ₂	- 0
(b)	N ₂	- 3
(c)	F ₂	- 1
(d)	O ₂	- 2

Q47: A, Phenyl methanamine

B. N,N-Dimethylaniline

C. N-Methyl aniline

D. Benzenamine

Choose the correct order of basic nature of the above amines.

(A) $D > B > C > A$

(B) $A > C > B > D$

(C) $A > B > C > D$

(D) $D > C > B > A$

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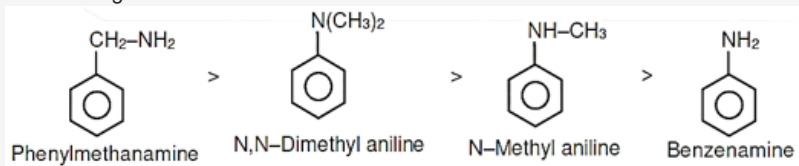
(B) $A > C > B > D$

(C) $A > B > C > D$

(D) $D > C > B > A$

Solution:

Basic strength order is



Q48: The nature of charge on resulting colloidal particles when FeCl_3 is added to excess of hot water is

- (A) Positive
- (B) Sometimes positive and sometimes negative
- (C) Neutral
- (D) Negative

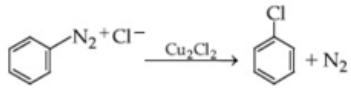
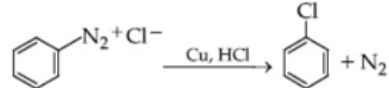
Q48: The nature of charge on resulting colloidal particles when FeCl_3 is added to excess of hot water is

- (A) Positive
- (B) Sometimes positive and sometimes negative
- (C) Neutral
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Solution:

If FeCl_3 is added to an excess of hot water, a positively charged sol of hydrated ferric oxide is formed due to the adsorption of Fe^{3+} ions.

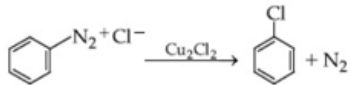
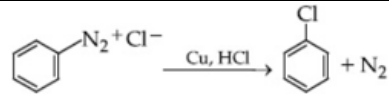
Q49: Match List-I with List-II

	List I		List II
A		(i)	Wurtz reaction
B		(ii)	Sandmeyer reaction
C	$2CH_3CH_2Cl + 2Na \xrightarrow{\text{Ether}} C_2H_5 - C_2H_5 + 2NaCl$	(iii)	Fittig reaction
D	$2C_6H_5Cl + 2Na \xrightarrow{\text{Ether}} C_6H_5 - C_6H_5 + 2NaCl$	(iv)	Gatterman reaction

Choose the correct answer from the options given below

- (A) A – (ii), B – (i), C – (iv), D – (iii)
- (B) A – (iii), B – (i), C – (iv), D – (ii)
- (C) A – (iii), B – (iv), C – (i), D – (ii)
- (D) A – (ii), B – (iv), C – (i), D – (iii)

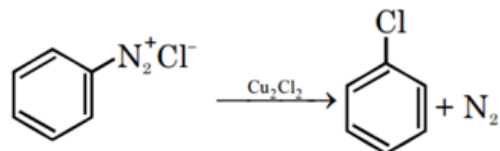
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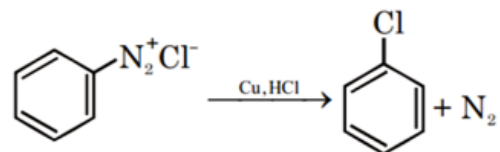
Choose the correct answer from the options given below

- (A) A – (ii), B – (i), C – (iv), D – (iii)
- (B) A – (iii), B – (i), C – (iv), D – (ii)
- (C) A – (iii), B – (iv), C – (i), D – (ii)
- (D) A – (ii), B – (iv), C – (i), D – (iii)

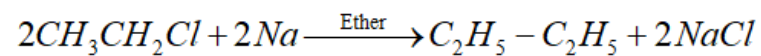
Solution:



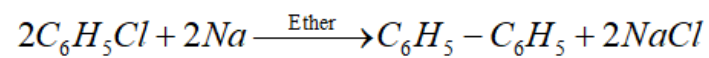
Sandmeyer's Reaction



Gattermann's Reaction



Wurtz reaction



Fittig reaction

Q50: Which of the following forms of hydrogen emits low energy β^- particles?

- (A) Protium ${}^1_1\text{H}$
- (B) Proton H^+
- (C) Tritium ${}^3_1\text{H}$
- (D) Deuterium ${}^2_1\text{H}$

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(A) Protium ${}^1_1\text{H}$

(B) Proton H^+

(C) Tritium ${}^3_1\text{H}$

(D) Deuterium ${}^2_1\text{H}$

Solution:

${}^3_1\text{H}$ (Tritium) is radioactive
 $\frac{n}{p}$ ratio is 2

Which is highly unstable so to reach stability range of $\frac{n}{p}$ 1 to 1.5, ${}^3_1\text{H}$ emits β^- particle



Q51: The pH ammonium phosphate solution, if pK_a of phosphoric acid and pK_b of ammonium hydroxide are 5.23 and 4.75 respectively, is _____(Round off to the Nearest Integer)

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7.00

Solution:

It is salt of WAWB

$$\begin{aligned} pH &= 7 + \frac{1}{2}pK_a - \frac{1}{2}pK_b \\ &= 7 + \frac{5.23}{2} - \frac{4.75}{2} \\ &= 7 + 2.615 - 2.375 \\ pH &= 7.24 \end{aligned}$$

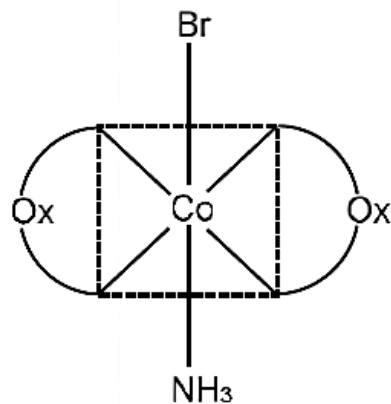
Ans. 7

Q52: The number of stereoisomers possible for $[\text{Co}(\text{ox})_2(\text{Br})(\text{NH}_3)]^{2-}$ is _____ [ox = oxalate]

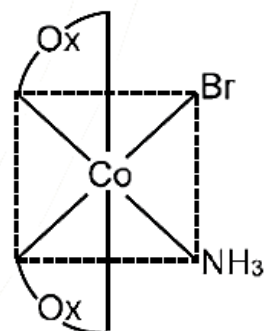
Q52: The number of stereoisomers possible for $[\text{Co}(\text{ox})_2(\text{Br})(\text{NH}_3)]^{2-}$ is _____ [ox = oxalate]

3.00

Solution:



Trans



Cis [d + ℓ]

Q53: Emf of the following cell at 298 K in V is $x \times 10^{-2}$.



The value of x is _____. (Rounded off to the nearest integer)

[Given: $E_{\text{Zn}^{2+}/\text{Zn}}^{\theta} = -0.76\text{V}$; $E_{\text{Ag}^+/\text{Ag}}^{\theta} = +0.80\text{V}$; $\frac{2.303RT}{F} = 0.059$]

Q53: Emf of the following cell at 298 K in V is $x \times 10^{-2}$.

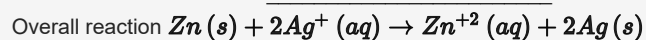
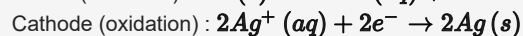
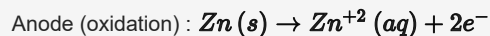


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147.00

Solution:



$$\begin{aligned} E_{\text{cell}} &= E_{\text{cell}}^{\theta} - \frac{0.059}{2} \log \frac{[\text{Zn}^{2+}]}{[\text{Ag}^+]^2} \\ &= (0.76 + 0.8) - \frac{0.059}{2} \log \frac{0.1}{(0.01)^2} \\ &= 1.56 - \frac{0.059}{2} \log 10^3 \\ &= 1.56 - \frac{0.059}{2} \times 3 \\ &= 1.56 - 0.09 \\ &= 1.47 \\ &= 147 \times 10^{-2} \text{V} \end{aligned}$$

Ans. 147

Q54: When 12.2 g of benzoic acid is dissolved in 100 g of water, the freezing point of solution was found to be -0.93°C ($K_f(\text{H}_2\text{O}) = 1.86 \text{ K kg mol}^{-1}$). The number (n) of benzoic acid molecules associated (assuming 100% association) is_____.

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2.00

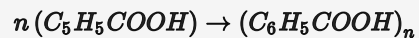
Solution:

$$\Delta T_f = i k_f m$$

$$0.93 = i \times 1.86 \left\{ \frac{12.2 \times 1000}{122 \times 100} \right\}$$

$$i = 0.5$$

There for benzoic acid associated as dimer.



$$i = 1 + \left(\frac{1}{n} - 1 \right) \beta \quad (\beta = 1) \text{ For 100\% association}$$

$$0.5 = \left(1 + \left(\frac{1}{n} - 1 \right) 1 \right)$$

So n = 2 Ans

Q55: If the activation energy of a reaction is 80.9 kJ mol^{-1} , the fraction of molecules at 700 K , having enough energy to react to form products is e^{-x} . The value of x is _____.

[Use $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$]

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[Use $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$]

14.00

Solution:

$$\text{Fraction of molecules crossing activation energy barrier} = e^{-\frac{E_a}{RT}}$$

$$= e^{-\frac{80.9 \times 10^3}{8.314 \times 700}}$$

$$= e^{-13.8}$$

$$\text{So } x = 13.8 \approx 14$$

Q56:

The average S–F bond energy in kJ mol^{-1} of SF_6 is _____. (Rounded off to the nearest integer)

[Given: The values of standard enthalpy of formation of $\text{SF}_6(\text{g})$, $\text{S}(\text{g})$ and $\text{F}(\text{g})$ are - 1100, 275 and 80 kJ mol^{-1} respectively.]

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The average S–F bond energy in kJ mol^{-1} of SF_6 is _____. (Rounded off to the nearest integer)

[Given: The values of standard enthalpy of formation of $\text{SF}_6(\text{g})$, $\text{S}(\text{g})$ and $\text{F}(\text{g})$ are - 1100, 275 and 80 kJ mol^{-1} respectively.]

309.00

Solution:

$$\begin{aligned} \text{SF}_6(\text{g}) &\rightarrow \text{S}(\text{g}) + 6\text{F}(\text{g}) \\ \Delta H_{\text{reaction}}^\circ &= 6(\text{BE}_{\text{S-F}}) = \Delta H_f^\circ(\text{F}, \text{g}) - \Delta H_f^\circ(\text{SF}_6, \text{g}) \\ &= 275 + 6 \times 80 - (-1100) \\ &= 1855 \end{aligned}$$

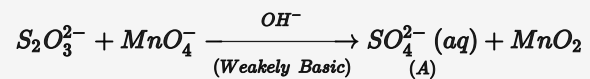
$$\text{BE}_{\text{S-F}} = \left(\frac{1855}{6} \right) = 309.17 \text{ kJ/mol} \approx 309 \text{ kJ/mol}$$

Q57: In a mildly alkaline medium, thiosulphate ion is oxidized by MnO_4^- to "A". The oxidation state of sulphur in "A" is _____.

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6.00

Solution:



So oxidation state of sulphur in product A is +6. Ans. 6

Q58: The NaNO_3 weighed out to make 50mL of an aqueous solution containing 70.0 mg Na^+ per mL is _____g. (Rounded off to the nearest integer)

[Given: Atomic weight in g mol^{-1} – Na; N:14; O:16]

Q58: The NaNO_3 weighed out to make 50mL of an aqueous solution containing 70.0 mg Na^+ per mL is _____g. (Rounded off to the nearest integer)

[Given: Atomic weight in g mol^{-1} – Na; N:14; O:16]

13.00

Solution:

$$[\text{Na}^+] = \frac{\frac{70 \times 10^{-3}}{23}}{\frac{1}{1000}} = \frac{70}{23} M$$

$$\text{Also, } [\text{Na}^+] = [\text{NaNO}_3] = \frac{70}{23} M$$

$$\text{Millimoles of } \text{NaNO}_3 = \frac{70}{23} \times 50$$

$$\text{Moles of } \text{NaNO}_3 = \frac{70}{23} \times 50 \times 10^{-3}$$

$$\text{Mass of } \text{NaNO}_3 (g) = \frac{70}{23} \times 50 \times 10^{-3} \times 85 = 12.93 g$$

Ans. 13

Q59: A ball weighing 10g is moving with a velocity of 90 ms^{-1} . If the uncertainty in its velocity is 5%, then the uncertainty in its position is _____ $\times 10^{-33}\text{m}$. (Rounded off to the nearest integer)

[Given: $h=6.63 \times 10^{-34}\text{Js}$]

Q59: A ball weighing 10g is moving with a velocity of 90 ms^{-1} . If the uncertainty in its velocity is 5%, then the uncertainty in its position is _____ $\times 10^{-33} \text{m}$. (Rounded off to the nearest integer)

[Given: $h=6.63 \times 10^{-34} \text{Js}$]

1.00

Solution:

$$\begin{aligned}\Delta v &= 90 \times \frac{5}{100} \\ &= 4.5 \text{ m/s} \\ \Delta v \cdot \Delta x &= \frac{h}{4\pi m} \\ \Delta x &= \frac{h}{4\pi m \cdot \Delta v} \\ &= \frac{6.63 \times 10^{-34}}{4 \times 3.14 \times 0.01 \times 4.5} \\ &= 1.17 \times 10^{-33}\end{aligned}$$

Q60: The number of octahedral voids per lattice site in a lattice is _____. (rounded off to the nearest integer)

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1.00

Solution:

For FCC unit cell $Z = 4$

$TV = 2(Z)$ and $OV = (Z)$

So, no. of octahedral void in FCC lattice = 4

So, Number of octahedral voids per lattice site in a lattice is = 1

Q61: If vectors $\vec{a}_1 = x\hat{i} - \hat{j} + \hat{k}$ and $\vec{a}_2 = \hat{i} + y\hat{j} + z\hat{k}$ are collinear, then a possible unit vector parallel to the vector $x\hat{i} + y\hat{j} + z\hat{k}$ is

(A) $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} - \hat{k})$

(B) $\frac{1}{\sqrt{2}}(\hat{i} - \hat{j})$

(C) $\frac{1}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$

(D) $\frac{1}{\sqrt{2}}(-\hat{j} + \hat{k})$

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(D) $\frac{1}{\sqrt{2}}(-\hat{j} + \hat{k})$

Solution:

Vectors $x\hat{i} - \hat{j} + \hat{k}$ and $\hat{i} + y\hat{j} + z\hat{k}$ are collinear.

$$\Rightarrow \frac{1}{x} = \frac{y}{-1} = \frac{z}{1}$$

$$\Rightarrow y = -z, x = \frac{1}{z}$$

$$\text{So, } x\hat{i} + y\hat{j} + z\hat{k} = \frac{1}{z}\hat{i} - z\hat{j} + z\hat{k}$$

Put $z = 1$, we get $\hat{i} - \hat{j} + \hat{k}$

A unit vector parallel to above vector is $\frac{1}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$

Q62: A seven digit number is formed using digits 3, 3, 4, 4, 4, 5, 5. The probability, that number so formed is divisible by 2, is

(A) $\frac{1}{7}$

(B) $\frac{6}{7}$

(C) $\frac{4}{7}$

(D) $\frac{3}{7}$

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(C) $\frac{4}{7}$

(D) $\frac{3}{7}$

Solution:

Numbers given are 3, 3, 4, 4, 4, 5, 5

Total number of 7 digit numbers = $\frac{7!}{2! \cdot 3! \cdot 2!}$

Number divisible by '2' has '4' at unit place

∴ Total favourable cases = $\frac{6!}{2! \cdot 2! \cdot 2!}$

∴ Required probability = $\frac{\frac{6!}{2! \cdot 2! \cdot 2!}}{\frac{7!}{2! \cdot 3! \cdot 2!}} = \frac{3}{7}$

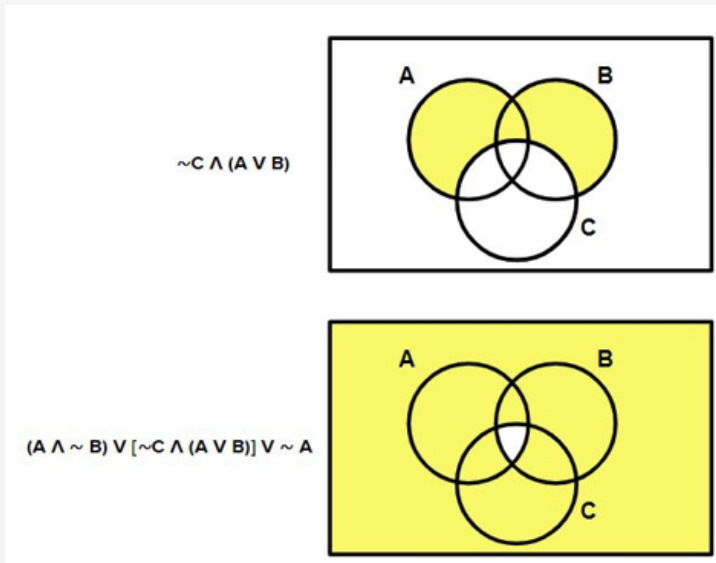
Q63: Let $F_1(A, B, C) = (A \wedge \sim B) \vee [\sim C \wedge (A \vee B)] \vee \sim A$ and $F_2(A, B) = (A \vee B) \vee (B \rightarrow \sim A)$ be two logical expressions. Then:

- (A) F_1 is not a tautology but F_2 is a tautology
- (B) Both F_1 and F_2 are not tautologies
- (C) F_1 is a tautology but F_2 is not a tautology
- (D) F_1 and F_2 both are tautologies

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- (D) F_1 and F_2 both are tautologies

Solution:



So, F_1 is not a Tautology.

$$F_2(A, B) = (A \vee B) \vee (\sim B \vee \sim A) = U$$

So, F_2 is a Tautology.

Q64: If the mirror image of the point $(1, 3, 5)$ with respect to the plane $4x - 5y + 2z = 8$ is (α, β, γ) , then $5(\alpha + \beta + \gamma)$ equals :

- (A) 39
- (B) 41
- (C) 47
- (D) 43

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(A) 39

(B) 41

(C) 47

(D) 43

Solution:

$$\frac{\alpha-1}{4} = \frac{\beta-3}{-5} = \frac{\gamma-5}{2} = \frac{-2(4-15+10-8)}{(16+25+4)} = \frac{2}{5}$$

$$\Rightarrow \alpha = \frac{13}{5}, \beta = 1, \gamma = \frac{29}{5}$$

$$\text{So, } 5(\alpha + \beta + \gamma) = 5\alpha + 5\beta + 5\gamma = 13 + 5 + 29 = 47$$

Q65: The triangle of maximum area that can be inscribed in a given circle of radius ' r ' is

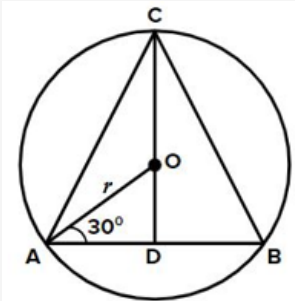
- (A) A right angle triangle having two of its sides of length $2r$ and r .
- (B) An equilateral triangle of height $\frac{2r}{3}$.
- (C) An isosceles triangle with base equal to $2r$.
- (D) An equilateral triangle having each of its side of length $\sqrt{3}r$.

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- (D) An equilateral triangle having each of its side of length $\sqrt{3}r$.

Solution:

Area of the inscribed triangle will be maximum in the case of symmetry i.e. when all the sides of the triangle are equal. So, triangle will be equilateral.



Side length of this equilateral triangle = $2AD = 2r \cos 30^\circ = \sqrt{3}r$

Q66: If $0 < a, b < 1$, and $\tan^{-1}a + \tan^{-1}b = \frac{\pi}{4}$, then the value of $(a + b) - \left(\frac{a^2+b^2}{2}\right) + \left(\frac{a^3+b^3}{3}\right) - \left(\frac{a^4+b^4}{4}\right) + \dots$ is:

- (A) $\log_e \left(\frac{e}{2}\right)$
- (B) $\log_e 2$
- (C) $e^2 - 1$
- (D) e

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(B) $\log_e 2$

(C) $e^2 - 1$

(D) e

Solution:

$$\tan^{-1}a + \tan^{-1}b = \frac{\pi}{4}$$

Apply $\tan$ both sides

$$\frac{a+b}{1-ab} = 1$$

$$a+b+ab=1$$

$$(1+a)(1+b)=2$$

$$\text{Now, } (a+b) - \left(\frac{a^2+b^2}{2}\right) + \left(\frac{a^3+b^3}{3}\right) - \left(\frac{a^4+b^4}{4}\right) + \dots$$

$$= \left(a - \frac{a^2}{2} + \frac{a^3}{3} - \frac{a^4}{4} + \dots\right) + \left(b - \frac{b^2}{2} + \frac{b^3}{3} - \frac{b^4}{4} + \dots\right)$$

$$= \log(1+a) + \log(1+b)$$

$$= \log[(1+a)(1+b)]$$

$$= \log 2$$

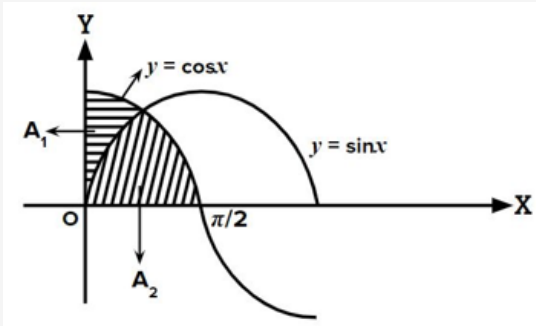
Q67: Let A_1 be the area of the region bounded by the curves $y = \sin x$, $y = \cos x$ and y -axis in the first quadrant. Also, let A_2 be the area of the region bounded by the curves $y = \sin x$, $y = \cos x$, x -axis and $x = \frac{\pi}{2}$ in the first quadrant. Then

- (A) $A_1 = A_2$ and $A_1 + A_2 = \sqrt{2}$
- (B) $A_1 : A_2 = 1 : \sqrt{2}$ and $A_1 + A_2 = 1$
- (C) $2A_1 = A_2$ and $A_1 + A_2 = 1 + \sqrt{2}$
- (D) $A_1 : A_2 = 1:2$ and $A_1 + A_2 = 1$

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- (C) $2A_1 = A_2$ and $A_1 + A_2 = 1 + \sqrt{2}$
- (D) $A_1 : A_2 = 1:2$ and $A_1 + A_2 = 1$

Solution:



$$A_1 + A_2 = \int_0^{\frac{\pi}{2}} \cos x \, dx = \sin x \Big|_0^{\frac{\pi}{2}} = 1$$

$$A_1 = \int_0^{\frac{\pi}{4}} (\cos x - \sin x) \, dx = (\sin x + \cos x) \Big|_0^{\frac{\pi}{4}} = \sqrt{2} - 1$$

$$\therefore A_2 = 1 - (\sqrt{2} - 1) = 2 - \sqrt{2}$$

$$\therefore \frac{A_1}{A_2} = \frac{\sqrt{2}-1}{2-\sqrt{2}} = 1 : \sqrt{2}$$

Q68: Let $f(x) = \sin^{-1}x$ and $g(x) = \frac{x^2-x-2}{2x^2-x-6}$. If $g(2) = \lim_{x \rightarrow 2} g(x)$, then the domain of the function $f \circ g$ is

- (A) $(-\infty, -1] \cup [2, \infty)$
- (B) $(-\infty, -2] \cup [-\frac{4}{3}, \infty)$
- (C) $(-\infty, -2] \cup [-1, \infty)$
- (D) $(-\infty, -2] \cup [-\frac{3}{2}, \infty)$

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- (D) $(-\infty, -2] \cup [-\frac{3}{2}, \infty)$

Solution:

$$g(2) = \lim_{x \rightarrow 2} \frac{(x-2)(x+1)}{(2x+3)(x-2)} = \frac{3}{7}$$

Clearly $f \circ g$ is defined for $x = 2$

If $x \neq 2$, then for $f \circ g(x)$ to be defined

$$x \neq -\frac{3}{2} \text{ and } \left| \frac{x^2-x-2}{2x^2-x-6} \right| \leq 1$$

$$x \neq -\frac{3}{2} \text{ and } \left| \frac{x+1}{2x+3} \right| \leq 1$$

$$x \neq -\frac{3}{2} \text{ and } (x+1)^2 \leq (2x+3)^2$$

$$x \neq -\frac{3}{2} \text{ and } (3x+4)(x+2) \geq 0$$

$$x \in (-\infty, -2] \cup \left[-\frac{4}{3}, \infty\right]$$

Hence, the domain of the function $f \circ g$ is

$$(-\infty, -2] \cup \left[-\frac{4}{3}, \infty\right)$$

Q69: Let $A(1, 4)$ and $B(1, -5)$ be two points. Let P be a point on the circle $(x - 1)^2 + (y - 1)^2 = 1$ such that $(PA)^2 + (PB)^2$ have maximum value, then the points, P , A and B lie on:

- (A) an ellipse
- (B) a hyperbola
- (C) a straight line
- (D) a parabola

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- (A) an ellipse
- (B) a hyperbola
- (C) a straight line
- (D) a parabola

Solution:

Let $P(1 + \cos \theta, 1 + \sin \theta)$

$$\therefore PA^2 = \cos^2 \theta + (\sin \theta - 3)^2 = 10 - 6 \sin \theta$$

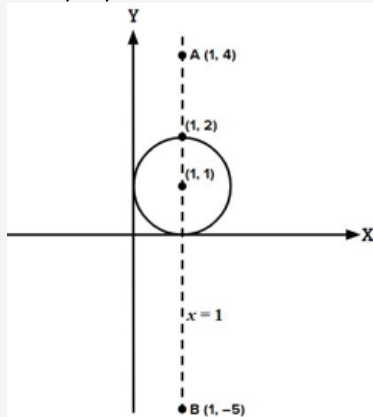
$$PB^2 = \cos^2 \theta + (\sin \theta + 6)^2 = 37 + 12 \sin \theta$$

$$PA^2 + PB^2 = 47 + 6 \sin \theta$$

It is maximum when $\theta = \frac{\pi}{2}$

$$\therefore P(1, 2)$$

$\therefore P, A, B$ lie on a line $x = 1$



Q70: If the locus of the mid-point of the line segment from the point $(3, 2)$ to a point on the circle, $x^2 + y^2 = 1$ is a circle of radius r , then r is equal to

(A) 1

(B) $\frac{1}{2}$

(C) $\frac{1}{4}$

(D) $\frac{1}{3}$

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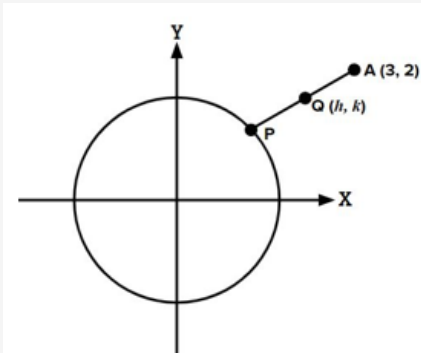
(A) 1

(B) $\frac{1}{2}$

(C) $\frac{1}{4}$

(D) $\frac{1}{3}$

Solution:



$$P \equiv (2h - 3, 2k - 2)$$

It lies on circle $x^2 + y^2 = 1$

$$\Rightarrow (2h - 3)^2 + (2k - 2)^2 = 1$$

$$\left(h - \frac{3}{2}\right)^2 + (k - 1)^2 = \frac{1}{4}$$

So, equation of locus is

$$\left(x - \frac{3}{2}\right)^2 + (y - 1)^2 = \frac{1}{4}$$

$$\Rightarrow \text{radius } r = \frac{1}{2}$$

Q71: Let $f(x) = \int_0^x e^t f(t) dt + e^x$ be a differentiable function for all $x \in \mathbb{R}$. Then $f(x)$

equals :

(A) $2e^{(e^x-1)} - 1$

(B) $e^{e^x} - 1$

(C) $e^{(e^x-1)}$

(D) $2e^{e^x} - 1$

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(B) $e^{e^x} - 1$

(C) $e^{(e^x-1)}$

(D) $2e^{e^x} - 1$

Solution:

$$f(x) = \int_0^x e^t f(t) dt + e^x$$

$$f'(x) = e^x f(x) + e^x$$

$$\frac{dy}{dx} = e^x (y + 1)$$

$$\int \frac{dy}{(y+1)} = \int e^x dx$$

$$\log|y+1| = e^x + C$$

$$y+1 = \pm e^C e^{e^x}$$

$$y+1 = k \cdot e^{e^x} \quad (\text{Put } \pm e^C = k)$$

$$\text{At } x=0, y=1$$

$$\Rightarrow 2 = ke$$

$$k = \frac{2}{e}$$

$$\Rightarrow y+1 = \frac{2e^{e^x}}{e}$$

$$\Rightarrow f(x) = 2e^{(e^x-1)} - 1$$

Q72: Consider the following system of equations:

$$x + 2y - 3z = a$$

$$2x + 6y - 11z = b$$

$$x - 2y + 7z = c,$$

where a , b and c are real constants. Then the system of equations:

- (A) has a unique solution when $5a = 2b + c$
- (B) has no solution for all a , b and c
- (C) has infinite number of solutions when $5a = 2b + c$
- (D) has a unique solution for all a , b and c

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- (D) has a unique solution for all a , b and c

Solution:

$$D = \begin{vmatrix} 1 & 2 & -3 \\ 2 & 6 & -11 \\ 1 & -2 & 7 \end{vmatrix}$$
$$= 20 - 2(25) - 3(-10)$$
$$= 20 - 50 + 30 = 0$$

$$D_1 = \begin{vmatrix} a & 2 & -3 \\ b & 6 & -11 \\ c & -2 & 7 \end{vmatrix}$$
$$= 20a - 2(7b + 11c) - 3(-2b - 6c)$$
$$= 20a - 14b - 22c + 6b + 18c$$
$$= 20a - 8b - 4c$$
$$= 4(5a - 2b - c)$$

$$D_2 = \begin{vmatrix} 1 & a & -3 \\ 2 & b & -11 \\ 1 & c & 7 \end{vmatrix}$$
$$= 7b + 11c - a(25) - 3(2c - b)$$
$$= 7b + 11c - 25a - 6c + 3b$$
$$= -25a + 10b + 5c$$
$$= -5(5a - 2b - c)$$

$$D_3 = \begin{vmatrix} 1 & 2 & a \\ 2 & 6 & b \\ 1 & -2 & c \end{vmatrix}$$
$$= 6c + 2b - 2(2c - b) - 10a$$
$$= -10a + 4b + 2c$$
$$= -2(5a - 2b - c)$$

For infinite solution

$$D = D_1 = D_2 = D_3 = 0$$
$$\Rightarrow 5a = 2b + c$$

Q73: The sum of the series $\sum_{n=1}^{\infty} \frac{n^2+6n+10}{(2n+1)!}$ is equal to

(A) $\frac{41}{8}e + \frac{19}{8}e^{-1} - 10$

(B) $-\frac{41}{8}e + \frac{19}{8}e^{-1} - 10$

(C) $\frac{41}{8}e - \frac{19}{8}e^{-1} - 10$

(D) $\frac{41}{8}e + \frac{19}{8}e^{-1} + 10$

Q73: The sum of the series $\sum_{n=1}^{\infty} \frac{n^2+6n+10}{(2n+1)!}$ is equal to

(A) $\frac{41}{8}e + \frac{19}{8}e^{-1} - 10$

(B) $-\frac{41}{8}e + \frac{19}{8}e^{-1} - 10$

(C) $\frac{41}{8}e - \frac{19}{8}e^{-1} - 10$

(D) $\frac{41}{8}e + \frac{19}{8}e^{-1} + 10$

Solution:

$$\sum_{n=1}^{\infty} \frac{n^2+6n+10}{(2n+1)!}$$

Put $2n + 1 = r$, where $r = 3, 5, 7, \dots$

$$\Rightarrow n = \frac{r-1}{2}$$

$$\frac{n^2+6n+10}{(2n+1)!} = \frac{\left(\frac{r-1}{2}\right)^2 + 3r - 3 + 10}{r!} = \frac{r^2 + 10r + 29}{4 \times r!}$$

$$\text{Now, } \sum_{r=3, 5, 7, \dots}^{\infty} \frac{r(r-1) + 11r + 29}{4 \times r!} = \frac{1}{4} \sum_{r=3, 5, 7, \dots}^{\infty} \left(\frac{1}{(r-2)!} + \frac{11}{(r-1)!} + \frac{29}{r!} \right)$$

$$= \frac{1}{4} \left\{ \left(\frac{1}{1!} + \frac{1}{3!} + \frac{1}{5!} + \dots \right) + 11 \left(\frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \dots \right) + 29 \left(\frac{1}{3!} + \frac{1}{5!} + \frac{1}{7!} + \dots \right) \right\}$$

$$= \frac{1}{4} \left\{ \frac{e - \frac{1}{e}}{2} + 11 \left(\frac{e + \frac{1}{e} - 2}{2} \right) + 29 \left(\frac{e - \frac{1}{e} - 2}{2} \right) \right\}$$

$$= \frac{1}{8} \left\{ e - \frac{1}{e} + 11e + \frac{11}{e} - 22 + 29e - \frac{29}{e} - 58 \right\}$$

$$= \frac{1}{8} \left\{ 41e - \frac{19}{e} - 80 \right\} = \frac{41}{8}e - \frac{19}{8}e^{-1} - 10$$

Q74: Let $f(x)$ be a differentiable function at $x = a$, such that $f'(a) = 2$, $f(a) = 4$. Then

$\lim_{x \rightarrow a} \frac{xf(a) - af(x)}{x - a}$ equals:

(A) $4 - 2a$

(B) $2a + 4$

(C) $2a - 4$

(D) $a + 4$

Q74: Let $f(x)$ be a differentiable function at $x = a$, such that $f'(a) = 2$, $f(a) = 4$. Then

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(A) $4 - 2a$

(B) $2a + 4$

(C) $2a - 4$

(D) $a + 4$

Solution:

On applying L-Hospital's Rule

$$\lim_{x \rightarrow a} \frac{xf(a) - af(x)}{x - a} = \lim_{x \rightarrow a} \frac{f(a) - af'(x)}{1} = f(a) - af'(a) = 4 - 2a$$

Q75: Let slope of the tangent line to a curve at any point $P(x, y)$ be given by $\frac{xy^2+y}{x}$. If the curve intersects the line $x + 2y = 4$ at $x = -2$, then the value of y , for which the point $(3, y)$ lies on the curve, is:

(A) $-\frac{4}{3}$

(B) $\frac{18}{35}$

(C) $-\frac{18}{11}$

(D) $-\frac{18}{19}$

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(A) $-\frac{4}{3}$

(B) $\frac{18}{35}$

(C) $-\frac{18}{11}$

(D) $-\frac{18}{19}$

Solution:

$$\frac{dy}{dx} = \frac{xy^2+y}{x}$$

$$\Rightarrow \frac{xdy-ydx}{y^2} = xdx$$

$$\Rightarrow -d\left(\frac{x}{y}\right) = d\left(\frac{x^2}{2}\right)$$

$$\Rightarrow \frac{-x}{y} = \frac{x^2}{2} + C$$

It passes through $(-2, 3)$

$$\Rightarrow \frac{2}{3} = 2 + C$$

$$\Rightarrow C = -\frac{4}{3}$$

$$\therefore \text{ curve is } \frac{-x}{y} = \frac{x^2}{2} - \frac{4}{3}$$

It also passes through $(3, y)$

$$\frac{-3}{y} = \frac{9}{2} - \frac{4}{3}$$

$$\Rightarrow \frac{-3}{y} = \frac{19}{6}$$

$$\Rightarrow y = -\frac{18}{19}$$

Q76: Let L be a line obtained from the intersection of two planes $x + 2y + z = 6$ and $y + 2z = 4$. If point $P(\alpha, \beta, \gamma)$ is the foot of perpendicular from (3, 2, 1) on L, then the value of $21(\alpha + \beta + \gamma)$ equals:

- (A) 68
- (B) 102
- (C) 142
- (D) 136

Q76: Let L be a line obtained from the intersection of two planes $x + 2y + z = 6$ and $y + 2z = 4$. If point $P(\alpha, \beta, \gamma)$ is the foot of perpendicular from (3, 2, 1) on L, then the value of $21(\alpha + \beta + \gamma)$ equals:

- (A) 68
- (B) 102
- (C) 142
- (D) 136

Solution:

A vector in the direction of line L is

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{vmatrix} = 3\hat{i} - 2\hat{j} + \hat{k}$$

So, D.R. of line L are $3, -2, 1$

A Point on line L is $(-2, 4, 0)$

So, equation of line L is

$$\frac{x+2}{3} = \frac{y-4}{-2} = \frac{z}{1}$$

Let $P(\alpha, \beta, \gamma) = (3\lambda - 2, -2\lambda + 4, \lambda)$ and $A(3, 2, 1)$

D.R. of AP are $3\lambda - 5, -2\lambda + 2, \lambda - 1$

$\therefore AP \perp \text{line L}$

$$\therefore 3(3\lambda - 5) - 2(-2\lambda + 2) + 1(\lambda - 1) = 0$$

$$\Rightarrow 14\lambda - 20 = 0$$

$$\Rightarrow \lambda = \frac{10}{7}$$

So, $P(\alpha, \beta, \gamma) = \left(\frac{16}{7}, \frac{8}{7}, \frac{10}{7}\right)$

$$\Rightarrow 21(\alpha + \beta + \gamma) = 21\left(\frac{34}{7}\right) = 102$$

Q77: For $x > 0$, if $f(x) = \int_1^x \frac{\log_e t}{(1+t)} dt$, then $f(e) + f\left(\frac{1}{e}\right)$ is equal to:

- (A) 1
- (B) $\frac{1}{2}$
- (C) 0
- (D) -1

Q77: For $x > 0$, if $f(x) = \int_1^x \frac{\log_e t}{(1+t)} dt$, then $f(e) + f\left(\frac{1}{e}\right)$ is equal to:

(A) 1

(B) $\frac{1}{2}$

(C) 0

(D) -1

Solution:

$$f(e) + f\left(\frac{1}{e}\right) = \int_1^e \frac{\log_e t}{1+t} dt + \int_1^{\frac{1}{e}} \frac{\log_e t}{1+t} dt = I_1 + I_2$$

$$I_2 = \int_1^{\frac{1}{e}} \frac{\log_e t}{1+t} dt$$

$$\text{Put } t = \frac{1}{z} \Rightarrow dt = -\frac{dz}{z^2}$$

$$I_2 = \int_1^e \frac{\log_e\left(\frac{1}{z}\right)}{1+\frac{1}{z}} \times \left(-\frac{dz}{z^2}\right) = \int_1^e \frac{\log_e z}{z(z+1)} dz = \int_1^e \frac{\log_e t}{t(t+1)} dt$$

$$f(e) + f\left(\frac{1}{e}\right) = \int_1^e \frac{\log_e t}{1+t} dt + \int_1^e \frac{\log_e t}{t(t+1)} dt = \int_1^e \left(\frac{\log_e t}{1+t} + \frac{\log_e t}{t(t+1)}\right) dt = \int_1^e \frac{\log_e t}{t} dt$$

$$\text{Put } \log t = u \Rightarrow \frac{1}{t} dt = du$$

$$f(e) + f\left(\frac{1}{e}\right) = \int_0^1 u du = \frac{u^2}{2} \Big|_0^1 = \frac{1}{2}$$

Q78: A natural number has prime factorization given by $n = 2^x 3^y 5^z$, where y and z are such that $y + z = 5$ and $y^{-1} + z^{-1} = \frac{5}{6}$, $y > z$. Then the number of odd divisors of n , including 1, is:

- (A) 6
- (B) $6x$
- (C) 11
- (D) 12

Q78: A natural number has prime factorization given by $n = 2^x 3^y 5^z$, where y and z are such that $y + z = 5$ and $y^{-1} + z^{-1} = \frac{5}{6}$, $y > z$. Then the number of odd divisors of n , including 1, is:

- (A) 6
- (B) $6x$
- (C) 11
- (D) 12

Solution:

$$y + z = 5 ; \frac{1}{y} + \frac{1}{z} = \frac{5}{6} \Rightarrow (y, z) = (2, 3) \text{ or } (3, 2)$$

$$\therefore \text{Number of odd divisors of } n = 2^x \cdot 3^y \cdot 5^z \text{ is } (y + 1)(z + 1) \\ = 3 \times 4 = 12$$

Q79: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} 2\sin\left(-\frac{\pi x}{2}\right), & \text{if } x < -1 \\ |ax^2 + x + b|, & \text{if } -1 \leq x \leq 1 \\ \sin(\pi x), & \text{if } x > 1 \end{cases}$$

If $f(x)$ is continuous on $\mathbb{R}$, then $a + b$ equals:

(A) -3

(B) 1

(C) 3

(D) -1

Q79: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} 2 \sin\left(-\frac{\pi x}{2}\right), & \text{if } x < -1 \\ |ax^2 + x + b|, & \text{if } -1 \leq x \leq 1 \\ \sin(\pi x), & \text{if } x > 1 \end{cases}$$

If $f(x)$ is continuous on $\mathbb{R}$, then $a + b$ equals:

(A) -3

(B) 1

(C) 3

(D) -1

Solution:

If f is continuous at $x = -1$, then

$$f(-1^-) = f(-1)$$

$$\Rightarrow 2 = |a - 1 + b|$$

$$\Rightarrow |a + b - 1| = 2 \quad \dots (i)$$

Similarly

$$f(1) = f(1^+)$$

$$\Rightarrow |a + b + 1| = 0$$

$$\Rightarrow a + b = -1$$

which also satisfies equation (i)

Q80: Let $A = \{1, 2, 3, \dots, 10\}$ and $f : A \rightarrow A$ be defined as

$$f(k) = \begin{cases} k+1 & \text{if } k \text{ is odd} \\ k & \text{if } k \text{ is even} \end{cases}$$

Then the number of possible functions $g : A \rightarrow A$ such that $g \circ f = f$ is

(A) 10^5

(B) ${}^{10}C_5$

(C) $5!$

(D) 5^5

Q80: Let $A = \{1, 2, 3, \dots, 10\}$ and $f : A \rightarrow A$ be defined as

$$f(k) = \begin{cases} k+1 & \text{if } k \text{ is odd} \\ k & \text{if } k \text{ is even} \end{cases}$$

Then the number of possible functions $g : A \rightarrow A$ such that $gof = f$ is

(A) 10^5

(B) ${}^{10}C_5$

(C) $5!$

(D) 5^5

Solution:

$$g(f(x)) = f(x)$$

$$\Rightarrow g(x) = x, \text{ when } x \text{ is even}$$

So total number of functions from A to A

$$= 10 \times 10 \times 10 \times 10 \times 10 = 10^5$$

Q81: Let a be an integer such that all the real roots of the polynomial $2x^5 + 5x^4 + 10x^3 + 10x^2 + 10x + 10$ lie in the interval $(a, a + 1)$. Then, $|a|$ is equal to _____.

Q81: Let a be an integer such that all the real roots of the polynomial $2x^5 + 5x^4 + 10x^3 + 10x^2 + 10x + 10$ lie in the interval $(a, a + 1)$. Then, $|a|$ is equal to _____.

2.00

Solution:

$$f(-1) = 3 > 0$$

$$f(-2) = -64 + 80 - 80 + 40 - 20 + 10 = -34 < 0$$

∴ Atleast one root in $(-2, -1)$

$$f'(x) = 10(x^4 + 2x^3 + 3x^2 + 2x + 1)$$

$$= 10x^2 \left(x^2 + \frac{1}{x^2} + 2 \left(x + \frac{1}{x} \right) + 3 \right)$$

$$= 10 \left(\left(x + \frac{1}{x} \right)^2 + 2 \left(x + \frac{1}{x} \right) + 1 \right)$$

$$= 10 \left(\left(x + \frac{1}{x} \right) + 1 \right)^2 > 0; \forall x \in R$$

∴ Exactly one real root in $(-2, -1)$

$$\Rightarrow a = -2 \Rightarrow |a| = 2$$

Q82: Let the normals at all the points on a given curve pass through a fixed point (a, b) . If the curve passes through $(3, -3)$ and $(4, -2\sqrt{2})$, and given that $a - 2\sqrt{2}b = 3$, then $(a^2 + b^2 + ab)$ is equal to _____.

Q82: Let the normals at all the points on a given curve pass through a fixed point (a, b) . If the curve passes through $(3, -3)$ and $(4, -2\sqrt{2})$, and given that $a - 2\sqrt{2}b = 3$, then $(a^2 + b^2 + ab)$ is equal to _____.

9.00

Solution:

Let the equation of normal is $Y - y = -\frac{1}{m}(X - x)$

Satisfy (a, b) in it

$$b - y = -\frac{1}{m}(a - x)$$

$$\Rightarrow b - y = -\frac{1}{\frac{dy}{dx}}(a - x)$$

$$\Rightarrow (b - y) dy = (x - a) dx$$

$$by - \frac{y^2}{2} = \frac{x^2}{2} - ax + c \quad \dots (i)$$

It passes through $(3, -3)$ and $(4, -2\sqrt{2})$

$$\therefore -3b - \frac{9}{2} = \frac{9}{2} - 3a + c$$

$$\Rightarrow -6b - 9 = 9 - 6a + 2c$$

$$\Rightarrow 6a - 6b - 2c = 18$$

$$\Rightarrow 3a - 3b - c = 9 \quad \dots (ii)$$

Also

$$-2\sqrt{2}b - 4 = 8 - 4a + c$$

$$4a - 2\sqrt{2}b - c = 12 \quad \dots (iii)$$

Also $a - 2\sqrt{2}b = 3 \quad \dots (iv)$ (given)

$$(ii) - (iii)$$

$$\Rightarrow -a + (2\sqrt{2} - 3)b = -3 \quad \dots (v)$$

$$(iv) + (v) \Rightarrow b = 0 \Rightarrow a = 3$$

$$\therefore a^2 + b^2 + ab = 9$$

Q83: Let $X_1, X_2, \dots, X_{18}$ be eighteen observations such that $\sum_{i=1}^{18} (X_i - \alpha) = 36$ and $\sum_{i=1}^{18} (X_i - \beta)^2 = 90$, where α and β are distinct real numbers. If the standard deviation of these observations is 1, then the value of $|\alpha - \beta|$ is _____.

Q83: Let $X_1, X_2, \dots, X_{18}$ be eighteen observations such that $\sum_{i=1}^{18} (X_i - \alpha) = 36$ and

$\sum_{i=1}^{18} (X_i - \beta)^2 = 90$, where α and β are distinct real numbers. If the standard deviation of

these observations is 1, then the value of $|\alpha - \beta|$ is _____.

4.00

Solution:

Observations	Mean
$X_i - \alpha$	2
X_i	$2 + \alpha$
$X_i - \beta$	$2 + \alpha - \beta$

Standard Deviation (X_i) = Standard Deviation ($X_i - \beta$)

$$\Rightarrow \frac{\sum_{i=1}^{18} (X_i - \beta)^2}{18} - \left[\frac{\sum_{i=1}^{18} (X_i - \beta)}{18} \right]^2 = 1$$

$$\Rightarrow \left[\frac{\sum_{i=1}^{18} (X_i - \beta)}{18} \right]^2 = 4$$

$$\Rightarrow \left| \frac{\sum_{i=1}^{18} (X_i - \beta)}{18} \right| = 2$$

$$\Rightarrow |2 + \alpha - \beta| = 2$$

$$\Rightarrow 2 + \alpha - \beta = 2 \text{ or } 2 + \alpha - \beta = -2$$

$$\Rightarrow \alpha - \beta = 0 \text{ or } \alpha - \beta = -4$$

$$\Rightarrow \alpha - \beta = -4 \quad (\because \alpha \neq \beta)$$

$$\Rightarrow |\alpha - \beta| = 4$$

Q84: Let α and β be two real numbers such that $\alpha + \beta = 1$ and $\alpha\beta = -1$. Let $p_n = (\alpha)^n + (\beta)^n$, $p_{n-1} = 11$ and $p_{n+1} = 29$ for some integer $n \geq 1$. Then, the value of p_n^2 is _____.

Q84: Let α and β be two real numbers such that $\alpha + \beta = 1$ and $\alpha\beta = -1$. Let $p_n = (\alpha)^n + (\beta)^n$, $p_{n-1} = 11$ and $p_{n+1} = 29$ for some integer $n \geq 1$. Then, the value of p_n^2 is _____.

324.00

Solution:

$$\begin{aligned}(\alpha + \beta) p_n &= (\alpha + \beta) (\alpha^n + \beta^n) \\(\alpha + \beta) p_n &= \alpha^{n+1} + \beta^{n+1} + \alpha\beta (\alpha^{n-1} + \beta^{n-1}) \\ \Rightarrow (1) p_n &= p_{n+1} - p_{n-1} = 29 - 11 = 18 \\ \Rightarrow p_n^2 &= 324\end{aligned}$$

Q85: If the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix}$ satisfies the equation

$A^{20} + \alpha A^{19} + \beta A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ for some real numbers α and β , then $\beta - \alpha$ is equal

to _____.

Q85: If the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix}$ satisfies the equation

$A^{20} + \alpha A^{19} + \beta A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ for some real numbers α and β , then $\beta - \alpha$ is equal

to _____.

4.00

Solution:

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 3 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 8 & 0 \\ 3 & 0 & -1 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

⋮
⋮
⋮
⋮

$$A^{19} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2^{19} & 0 \\ 3 & 0 & -1 \end{bmatrix}, \quad A^{20} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2^{20} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Given,

$$A^{20} + \alpha A^{19} + \beta A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 + \alpha + \beta & 0 & 0 \\ 0 & 2^{20} + \alpha 2^{19} + 2\beta & 0 \\ 3\alpha + 3\beta & 0 & 1 - \alpha - \beta \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \alpha + \beta = 0 \text{ and } 2^{20} + \alpha 2^{19} + 2\beta = 4$$

$$\Rightarrow 2^{20} + \alpha (2^{19} - 2) = 4$$

$$\Rightarrow \alpha = \frac{4 - 2^{20}}{2^{19} - 2} = -2$$

$$\Rightarrow \beta = 2 \Rightarrow (\beta - \alpha) = 4$$

Q86: Let L be a common tangent line to the curves $4x^2 + 9y^2 = 36$ and $(2x)^2 + (2y)^2 = 31$.

Then the square of the slope of the line L is _____.

Q86: Let L be a common tangent line to the curves $4x^2 + 9y^2 = 36$ and $(2x)^2 + (2y)^2 = 31$.

Then the square of the slope of the line L is _____.

3.00

Solution:

Given curves are $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and $x^2 + y^2 = \frac{31}{4}$

Let equation of their common tangent with slope m is

$$y = mx + c$$

$$\therefore c^2 = 9m^2 + 4 = \frac{31}{4}m^2 + \frac{31}{4}$$

$$\Rightarrow \frac{5m^2}{4} = \frac{15}{4}$$

$$\Rightarrow m^2 = 3$$

Q87: Let z be those complex numbers which satisfy

$$|z + 5| \leq 4 \text{ and } z(1 + i) + \bar{z}(1 - i) \geq -10, \quad i = \sqrt{-1}.$$

If the maximum value of $|z + 1|^2$ is $\alpha + \beta\sqrt{2}$, then the value of $(\alpha + \beta)$ is _____.

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$$|z + 5| \leq 4 \text{ and } z(1 + i) + \bar{z}(1 - i) \geq -10, \quad i = \sqrt{-1}.$$

If the maximum value of $|z + 1|^2$ is $\alpha + \beta\sqrt{2}$, then the value of $(\alpha + \beta)$ is _____.

48.00

Solution:

$|z + 5| \leq 4$ represents a circle and its interior having centre $(-5, 0)$ and radius 4

Also given,

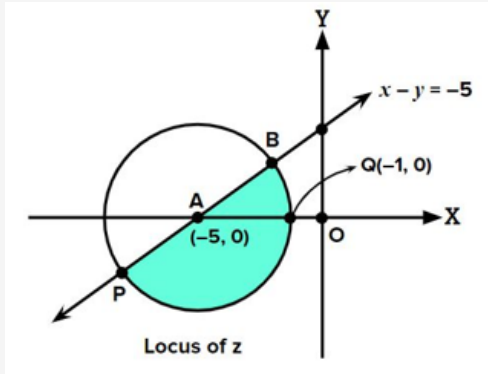
$$z(1 + i) + \bar{z}(1 - i) \geq -10$$

$$\Rightarrow (z + \bar{z}) + i(z - \bar{z}) \geq -10$$

$$(Put \ z = x + iy \Rightarrow \bar{z} = x - iy)$$

$$\Rightarrow 2x + i(2iy) \geq -10$$

$$\Rightarrow x - y \geq -5$$



$$P \equiv (-5 - 4 \cos 45^\circ, 0 - 4 \sin 45^\circ) = (-5 - 2\sqrt{2}, -2\sqrt{2})$$

$$|z + 1|_{\max}^2 = (PQ)^2 = 32 + 16\sqrt{2}$$

$$\alpha = 32$$

$$\beta = 16$$

$$\alpha + \beta = 48$$

Q88: If $I_{m,n} = \int_0^1 x^{m-1} (1-x)^{n-1} dx$, for $m, n \geq 1$ and

$\int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx = \alpha I_{m,n}$, $\alpha \in \mathbb{R}$, then α equals _____.

Q88: If $I_{m,n} = \int_0^1 x^{m-1} (1-x)^{n-1} dx$, for $m, n \geq 1$ and

$\int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx = \alpha I_{m,n}$, $\alpha \in R$, then α equals _____.

1.00

Solution:

$$I_{m,n} = \int_0^1 x^{m-1} \cdot (1-x)^{n-1} dx$$

$$\text{Put } x = \frac{y}{y+1}$$

$$I_{m,n} = \int_0^1 \frac{y^{m-1}}{(y+1)^{m+n}} (-1) dy = \int_1^\infty \frac{y^{m-1}}{(y+1)^{m+n}} dy \quad \dots (i)$$

$$\text{Similarly from, } I_{m,n} = \int_0^1 x^{n-1} \cdot (1-x)^{m-1} dx,$$

$$I_{m,n} = \int_0^1 \frac{y^{n-1}}{(y+1)^{m+n}} dy \quad \dots (ii)$$

$$(i) + (ii)$$

$$2I_{m,n} = \int_0^\infty \frac{y^{m-1} + y^{n-1}}{(y+1)^{m+n}} dy$$

$$\Rightarrow 2I_{m,n} = \int_0^1 \frac{y^{m-1} + y^{n-1}}{(y+1)^{m+n}} dy + \int_1^\infty \frac{y^{m-1} + y^{n-1}}{(y+1)^{m+n}} dy$$

$$\text{Consider } \int_1^\infty \frac{y^{m-1} + y^{n-1}}{(y+1)^{m+n}} dy$$

$$\text{Put } y = \frac{1}{z}$$

$$\Rightarrow \int_1^\infty \frac{y^{m-1} + y^{n-1}}{(y+1)^{m+n}} dy = \int_1^0 \frac{z^{m+1} + z^{n+1}}{(z+1)^{m+n}} \frac{(-1)}{z^2} dz = \int_0^1 \frac{z^{m-1} + z^{n-1}}{(z+1)^{m+n}} dz = \int_0^1 \frac{y^{m-1} + y^{n-1}}{(y+1)^{m+n}} dy$$

$$\Rightarrow 2I_{m,n} = 2 \int_0^1 \frac{y^{m-1} + y^{n-1}}{(y+1)^{m+n}} dy$$

$$\Rightarrow I_{m,n} = \int_0^1 \frac{y^{m-1} + y^{n-1}}{(y+1)^{m+n}} dy = \int_0^1 \frac{x^{m-1} + x^{n-1}}{(1+x)^{m+n}} dx \Rightarrow \alpha = 1$$

Q89: The total number of 4-digit numbers whose greatest common divisor with 18 is 3, is

Q89: The total number of 4-digit numbers whose greatest common divisor with 18 is 3, is

1000.00

Solution:

Number of required numbers

$=$ 4 digit numbers divisible by 3 $-$ 4 digit numbers divisible by 6 $-$ 4 digit numbers divisible by 9 $+$ 4 digit numbers divisible by 18

$= 3000 - 1500 - 1000 + 500$

$= 1000$

Q90: If the arithmetic mean and geometric mean of the p^{th} and q^{th} terms of the sequence $-16, 8, -4, 2, \dots$ satisfy the equation $4x^2 - 9x + 5 = 0$, then $p + q$ is equal to

Q90: If the arithmetic mean and geometric mean of the p^{th} and q^{th} terms of the sequence $-16, 8, -4, 2, \dots$ satisfy the equation $4x^2 - 9x + 5 = 0$, then $p + q$ is equal to

10.00

Solution:

Given sequence is $-16, 8, -4, 2, \dots$

It is a GP with common ratio $r = -\frac{1}{2}$

Its n^{th} term is $a_n = (-16) \left(-\frac{1}{2}\right)^{n-1}$

Roots of $4x^2 - 9x + 5 = 0$ are $1, \frac{5}{4}$

$\therefore GM \leq AM \therefore GM = 1$

Now GM of p^{th} and q^{th} term $= \sqrt{(-16) \left(-\frac{1}{2}\right)^{p-1} \cdot (-16) \left(-\frac{1}{2}\right)^{q-1}}$

$$\Rightarrow 16 \left(-\frac{1}{2}\right)^{\frac{p+q-2}{2}} = 1$$

$$\Rightarrow \left(-\frac{1}{2}\right)^{\frac{p+q-2}{2}} = \frac{1}{16}$$

$$\Rightarrow \frac{p+q-2}{2} = 4$$

$$\Rightarrow p + q = 10$$